Center-Pixel Guided Dual-Path Multihead Attention Fusion Transformer for Hyperspectral Image Classification

Haobo Jiang, Qiankun Gao, Haobing Chen, and Bin Zhou

***Abstract*—The performance of hyperspectral image classification (HSIC) relies on the deep integration of spatial and spectral information. Although mainstream Transformer architectures can capture global context, they often overlook the key inductive bias that "the central pixel determines the category," and also incur high computational costs. Moreover, traditional linear fusion strategies (such as concatenation or element-wise addition) struggle to capture the complex nonlinear interactions between spatial and spectral dimensions. To address these issues, this article proposes a central-pixel guided dual-path multihead attention transformer (CPFormer). The model first introduces a spectral–spatial feature embedding (SSFE) module, which generates robust initial representations while preserving inherent spectral characteristics. Guided by the central pixel-guided dual-path multihead attention (CP-DMA) module is then proposed: the spatial path uses the central pixel as the query and incorporates 2-D Gaussian positional encoding to enhance local perception; the spectral path employs depthwise separable convolution and bidirectional spectral decay prior to model spectral (channel) continuity. Furthermore, CP-DMA adopts element-wise multiplication for spatial–spectral fusion, enabling adaptive and nonlinear feature interaction. Information plane analysis further confirms that, compared with linear fusion, the multiplication strategy can more effectively compress redundant information and drive feature representations closer to the "minimum sufficient statistic" in information theory. Extensive experiments on four benchmark datasets under limited training samples demonstrate that CPFormer outperforms state-of-the-art methods in classification performance. The source code will be available at: https://github.com/Jianghaob/CPFormer**

***Index Terms*—Center-pixel guidance, dual-path attention, hyperspectral image classification (HSIC), multiplicative fusion.**

I. Introduction

HYPERSPECTRAL imaging is a pivotal technology in the field of remote sensing. Unlike conventional RGB images that contain only three channels, hyperspectral images (HSIs) typically encompass tens to hundreds of narrow, contiguous spectral bands. This rich spectral information endows them with unique advantages in precise target identification [1], [2], and detection [3], [4], enabling widespread applications in precision agriculture [5], environmental monitoring [6], and mineral exploration [7]. HSI classification (HSIC), as the fundamental task underlying these applications, aims to assign a unique land-cover class label to each pixel in the image, with its classification accuracy directly influencing the reliability of downstream tasks.

Over the past decades, HSIC methods have undergone a profound evolution from traditional machine learning (ML-based) to deep learning (DL-based) approaches. Early research primarily relied on dimensionality reduction techniques such as principal component analysis (PCA) [8] and linear discriminant analysis (LDA) [9] combined with classifiers like support vector machines (SVM) [10]. The emergence of DL brought a paradigm shift, with convolutional neural networks (CNNs) [11], [12], graph neural networks (GNNs) [13], [14], Transformers [15], [16], [17], and state space models (SSMs) [18], [19], [20] successively emerging to continuously advance classification performance from different perspectives including local feature extraction, irregular structure modeling, global dependence capture, and efficient sequence processing.

However, current mainstream HSIC methods predominantly adopt the patch-based classification paradigm, where the class label of the central pixel serves as the classification target for the entire image patch, with neighboring pixels functioning solely as spatial context for auxiliary discrimination. Although the aforementioned DL architectures demonstrate exceptional performance in general vision tasks [15], [21], they were not originally designed for this "center-guided" characteristic: CNNs gradually expand the receptive field through stacked convolutional layers, causing contributions from different spatial positions to become equalized. The message passing mechanism of GNNs aggregates neighboring nodes uniformly. The global self-attention of Transformers treats all tokens equally [see Fig. 1(a)] and the sequential scanning of SSMs lacks an explicit spatial hierarchical structure. This mismatch between these methods and the requirements of the patch-based paradigm may cause noise features from boundary regions or heterogeneous neighborhoods to dilute the discriminative information of the central pixel.

To address these issues, researchers have conducted extensive explorations. Early works introduced central attention modules (CAMs) within CNN frameworks [23], [24], [25], achieving adaptive weighting by computing similarities

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The authors are with the School of Computer Science and the School of Artificial Intelligence, South–Central Minzu University, Wuhan, Hubei 430000, China (e-mail: hbjiang@scuec.edu.cn; qkgao@scuec.edu.cn; hbchen@scuec.edu.cn; binzhou@mail.scuec.edu.cn).

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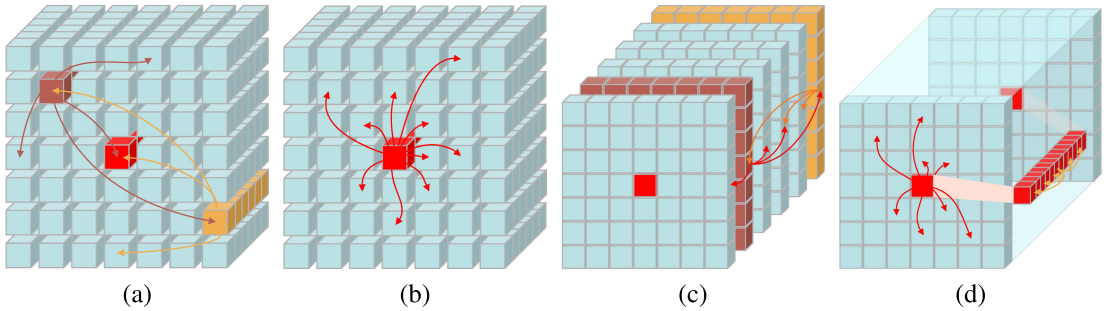


Fig. 1. Comparison of different attention mechanisms. (a) Standard self-attention in the spatial domain. (b) Center-pixel guided SA. (c) Standard self-attention in the spectral domain. (d) Center-pixel guided spectral–spatial dual-path attention.

between central and neighboring pixels. In recent years, with the widespread adoption of Transformers in HSIC, methods such as CSIL [26], CAT [27], and CP-Transformer [28] have combined the central guidance concept with self-attention mechanisms, achieving significant progress. However, in-depth analysis reveals that existing methods still exhibit the following limitations: First, the guidance dimension is singular. Existing mechanisms predominantly focus on central guidance in the spatial dimension [see Fig. 1(b)] while neglecting collaborative modeling in the spectral dimension. For instance, CAT [27] only introduces Gaussian distance encoding in spatial attention (SA). Second, the fusion strategy is delayed. Features from spatial and spectral branches are typically fused linearly through concatenation or addition at the network's end [26], [29], lacking deep interaction at early stages, which limits the full utilization of complementary spatial–spectral information. Third, physical prior constraints are insufficient. Attention weights mostly rely on purely data-driven learning without explicitly incorporating physical properties of hyperspectral data (such as spatial distance decay and spectral continuity) as inductive biases.

To address the aforementioned issues, this article proposes the central pixel-guided dual-path multihead attention fusion Transformer (CPFormer). The core idea is to reconstruct the traditional Transformer's "all-to-all" attention interaction into a "one-to-all" directed aggregation anchored on the central pixel, implemented in parallel across both spatial and spectral orthogonal dimensions [see Fig. 1(d)]. In particular, this article designs the central pixel-guided dual-path multihead attention (CP-DMA) module as the core component: the spatial path employs the central pixel feature as the sole query vector, explicitly modeling spatial distance decay priors through 2-D Gaussian positional encoding; the spectral path unfolds each channel of the central pixel as a query sequence, leveraging bidirectional spectral decay positional encoding to strengthen correlations among adjacent channels. The outputs from both paths undergo element-wise multiplication at each attention head level, achieving early-stage nonlinear fusion and adaptive calibration of spatial–spectral features. By stacking multiple Transformer encoders based on CP-DMA, the model progressively refines the center-guided joint spatial–spectral representation layer by layer.

The main contributions of this article are summarized as follows.

1) This article proposes the CPFormer architecture. Targeting the problem that existing methods lack explicit modeling of the central pixel in patch-based HSIC, we reconstruct the standard self-attention's "all-to-all" computation into a "one-to-all" directed aggregation anchored on the central pixel, ensuring focus on classification-critical features at the mechanism level while reducing attention computation complexity from $O(N^2)$ to $O(N)$.
2) This article designs the CP-DMA dual-path attention module. We simultaneously construct central-guided parallel attention mechanisms in both spatial and spectral dimensions, introducing Gaussian positional encoding and bidirectional decay encoding as physical prior constraints, respectively. Element-wise multiplication is employed for each head to achieve early-stage nonlinear fusion of spatial–spectral features, overcoming the expressive bottleneck of traditional late-stage linear fusion.
3) Experiments on multiple public HSI benchmark datasets demonstrate that CPFormer outperforms current state-of-the-art methods in terms of classification accuracy and robustness.

II. Related Work

This section first reviews the evolution of representative methods in the HSIC field, then focuses on research related to the central pixel guidance mechanism that is closely relevant to this article.

A. HSI Classification

HSIC can be broadly categorized into ML-based and DL-based approaches, where DL methods can be further divided into CNN-based, GNN-based, Transformer-based, and SSM-based methods.

1) ML-Based Methods: Early methods primarily focused on feature engineering and classifier design. To alleviate the "Hughes phenomenon" [30] caused by high-dimensional spectral data, researchers extensively employed dimensionality reduction techniques such as PCA [8], LDA [9], and band selection [31], combined with classifiers including SVM [10] and random forests (RFs) [32] for classification. To enhance

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spatial modeling capability, techniques such as extended morphological profiles (EMPs) [33], local binary patterns (LBPs) [34], and superpixel segmentation [35] were introduced to construct joint spatial–spectral features. These methods established an important foundation for HSIC. However, the expressive capacity of handcrafted features is limited, making it difficult to capture deep nonlinear structures in the data. The emergence of DL methods brought a paradigm shift.

2) CNN-Based Methods: CNNs became an early research focus owing to their advantages in local feature extraction and parameter sharing. Yang et al. [11] proposed a dual-branch CNN to jointly extract spatial–spectral features. Roy et al. [12] introduced HybridSN, which fuses 3-D and 2-D convolutions and has become a classic baseline. Subsequent research has continuously optimized performance and efficiency through strategies such as adaptive convolutional kernels [36], multiscale pyramids [37], and depthwise separable convolutions [25].

3) GNN-Based Methods: GNNs, represented by graph convolutional networks (GCNs), have expanded the research frontier with their flexible topology modeling capabilities. GCN achieves flexible representation of irregular spatial structures by modeling pixels as graph nodes [38], [39]. Shahraki and Prasad [40] fused 1D-CNN with GCN for HSIC. Hong et al. [41] proposed miniGCN, which alleviates memory pressure through mini-batch training. Wang et al. [42] constructed a superpixel-pixel dual-branch network to achieve multiscale feature collaborative extraction. However, GCN faces challenges, including high graph construction complexity and substantial adjacency matrix storage overhead, limiting its application to large-scale data.

4) Transformer-Based Methods: Transformers [15] have brought new breakthroughs to HSIC through the global modeling capability of self-attention mechanisms. SpectralFormer [16] and SSFTT [17] pioneered the combination of spectral sequence processing with spatial token modeling, demonstrating superior performance over traditional CNNs and GCNs. Subsequent research has continuously improved performance through strategies such as convolutional inductive biases [43], multibranch fusion [44], [45], cross-scale attention [46], [47], and information enhancement [48]. In addition, SegHSI [49] explored efficient semantic segmentation paradigms, while SpectralX [50] and BiDA [51] proposed effective solutions for domain generalization and label-limited scenarios. Concurrently, UniTS [52] extends Transformers to the unified modeling of diverse remote sensing tasks. Nevertheless, the global self-attention mechanism of Transformers introduces $O(N^2)$ computational complexity and treats all pixels equally, lacking explicit modeling of spatial structure.

5) SSM-Based Methods: Recently, SSMs represented by Mamba [19] have emerged as a powerful complement to Transformers with their linear-complexity sequence modeling capability [18]. SpectralMamba [20] first introduced SSMs to HSIC, addressing spectral variability through dynamic masking and segmented scanning. Subsequent works designed decoupled spatial–spectral structures [53] and 3-D state transition mechanisms [54]. MambaHSI [55] achieved the first image-level global modeling.

In summary, the aforementioned methods have advanced HSIC technology from different perspectives. However, under the patch-based classification paradigm, their processing of pixels within image patches tends toward equalization, failing to fully embody the core prior that "the central pixel determines the classification label."

B. Central Pixel Guidance and Attention Mechanisms

To better adapt models to the intrinsic characteristics of patch-based classification, researchers have proposed a series of attention mechanisms centered on the central pixel.

Early explorations within the CNN framework established an important foundation. Zhao et al. [23] embedded a CAM after 3-D convolutional features, achieving adaptive weighted aggregation by computing correlations between neighborhood and central features. Liu et al. [24] proposed the CAN model, its simplified variant (miniCAN), systematically articulating the feature extraction principle that "category is determined by the central spectrum, and effective spatial information originates from similar neighborhoods." Li et al. [25] proposed the CVSSN, which hierarchically exploits center-oriented self-similarity relationships within both the input and feature spaces. Similarly, Zhao et al. [56] introduced the CG-SA module, which fuses cosine similarity with Gaussian-Euclidean similarity to more intricately characterize the correlations between the central pixel and its neighbors. These works validated the effectiveness of the central guidance strategy.

In-depth research within the Transformer framework has further expanded the design space. Yang et al. [26] proposed the CSIL model, which enhances fine-grained feature extraction and long-short range dependence modeling via hierarchical sampling and a dedicated center-surround Transformer; however, the features from its dual branches are merely fused via concatenation at the late stage. Liu et al. [29] developed the lightweight MATA model, which performs multiregion sampling around the central pixel to selectively capture features from the target and similar regions, thereby effectively mitigating background interference. In addition, CAT [27] introduced Gaussian positional encodings to assign attention weights based on spatial distance, explicitly integrating physical priors into the center-guided mechanism, yet this design was limited solely to the spatial dimension. Chen et al. [57] proposed a "spectral query spatial" paradigm utilizing relative positional encodings to guide central focus. Furthermore, Yu et al. [28] introduced the CP-Transformer, which employs the central feature as the sole query vector to enhance class discriminability while reducing computational complexity. Nevertheless, its single-path design restricts the synergistic exploitation of joint spectral–spatial information.

However, existing central guidance methods still have room for improvement in three aspects: first, they predominantly focus on a single spatial or spectral dimension, lacking collaborative design across both dimensions; second, spatial–spectral features are mostly fused linearly at late network stages with insufficient early interaction; third, the incorporation of physical priors remains inadequate. The CP-DMA module proposed in this article is specifically designed to address these limitations: it simultaneously constructs central guidance

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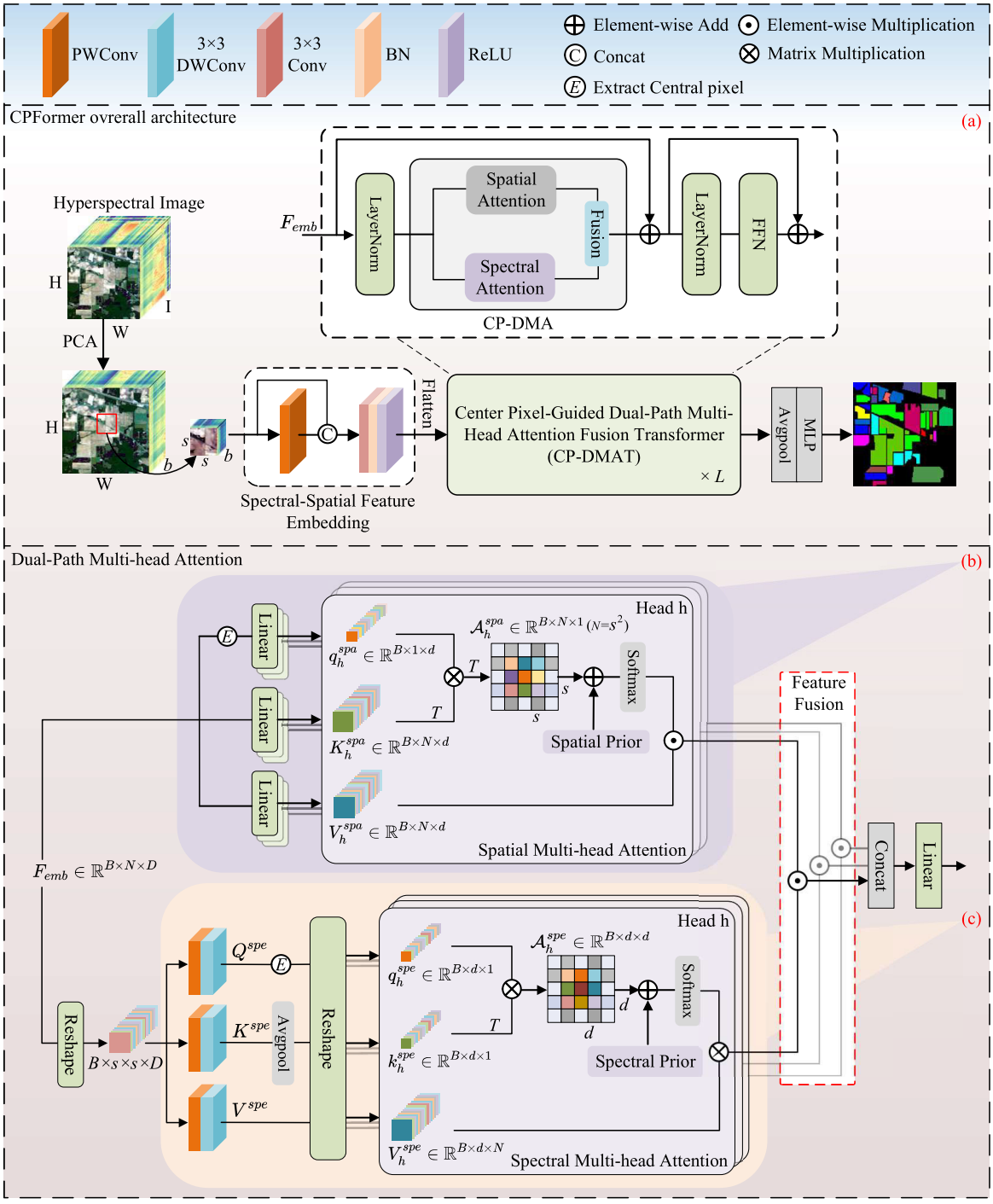


Fig. 2. Overall architecture and details of the proposed CPFormer. (a) CPFormer overall architecture. (b) Spatial MHA in the CP-DMA. (c) Spectral MHA in the CP-DMA.

mechanisms in both spatial and spectral dimensions, introduces Gaussian encoding and bidirectional decay encoding as physical constraints, respectively, and achieves early-stage nonlinear fusion of spatial–spectral features through element-wise multiplication.

III. Methodology

A. Overview

The proposed CPFormer architecture is illustrated in Fig. 2(a). The architecture comprises three core components: a spectral–spatial feature embedding (SSFE) module that achieves effective feature transformation while preserving original spectral information; a CP-DMA transformer encoder (CP-DMAT) that models contextual dependencies from both spatial and spectral orthogonal dimensions; and an MLP classifier for final class prediction.

Given an original HSI $\mathbf{X} \in \mathbb{R}^{H\times W\times I}$, where H, W, and I denote the height of spatial dimension, the width of spatial dimension, and the number of spectral bands, respectively. The ground-truth class label for each pixel is denoted as $Y \in \{1, 2, \ldots, C\}$, where C represents the total number of classes. To mitigate the Hughes phenomenon [30] and

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suppress noise interference, we first apply PCA to reduce the dimensionality to $\mathbf{X}_{\text{pca}} \in \mathbb{R}^{H\times W\times b}$, where $b \ll I$ represents the number of principal components (PCs) retained. It is worth noting that PCA, as a linear orthogonal transformation method, generates PCs that are linear combinations of the original spectral bands and are mutually uncorrelated. However, these PCs effectively constitute a latent spectral space containing the most discriminative information. Therefore, following [12], [17], [44], [48], this article treats the channel dimension of $\mathbf{X}_{\text{pca}}$ as synonymous with the "spectral dimension," both referring to this transformed feature space. Subsequently, for each pixel to be classified, we extract a 3-D neighborhood patch of spatial size $s \times s$ as $\mathbf{P} \in \mathbb{R}^{s\times s\times b}$ to represent the local context. To ensure that edge pixels can generate complete patches, we apply mirror reflection padding with width $(s-1)/2$ to $\mathbf{X}_{\text{pca}}$. The class label of each patch is determined by its central pixel, a principle that permeates the entire method design and guides the construction of the attention mechanism. After generating all patches, we remove patches with background labels (i.e., $Y = 0$) and divide them into training and test sets according to predetermined quantities.

B. Spectral–Spatial Feature Embedding

As a critical bridge connecting the raw HSI input and the Transformer, the design quality of the feature embedding module directly impacts the effectiveness of subsequent feature learning. Existing methods either adopt linear projection [16], [28] after flattening to generate a single feature representation, relegating the task of learning complex spectral–spatial interactions entirely to subsequent stages; or employ shallow convolutions (e.g., cascaded 3-D–2-D architectures [17], [43], [44], [45], [47]), which can introduce spatial priors but tend to cause premature fusion of spectral–spatial information. Given the core objective of this article—realizing spectral–spatial feature interaction based on center-pixel guidance—a feature that is excessively fused at the input stage or an overly simplistic feature is unsuitable as the input to the subsequent Transformer encoder. Therefore, we adopt a compromise strategy and design a three-stage "convolution-concatenation-convolution" feature embedding module (related experiments are shown in Section IV-G1), aiming to generate feature representations with preliminary spectral enhancement and local spatial awareness. In particualr, we first employ 1×1 convolution to learn initial interband correlation features $\mathbf{F}_{\text{init}}$ while keeping the spatial structure unchanged; then concatenate the original input $\mathbf{P}$ with $\mathbf{F}_{\text{init}}$ along the channel dimension to preserve the original spectral features; finally, we use 3×3 convolution to capture local spatial dependencies, and combine batch normalization (BN) and ReLU activation to achieve nonlinear feature transformation, yielding the final embedded features $\mathbf{F}_{\text{emb}}$. The 2-D feature map $\mathbf{F}_{\text{emb}}$ is then flattened into a token sequence $\mathbf{T}$ (where $N = s^2$), which serves as the input to the subsequent Transformer encoder. This process can be formulated as follows:

$$\begin{aligned}\mathbf{F}_{\text{init}} &= \text{Conv}_{1\times 1}(\mathbf{P}) \in \mathbb{R}^{B\times s\times s\times b}\\ \mathbf{F}_{\text{cat}} &= \text{Concat}([\mathbf{P}, \mathbf{F}_{\text{init}}]) \in \mathbb{R}^{B\times s\times s\times 2b}\\ \mathbf{F}_{\text{emb}} &= \text{ReLU}\left(\text{BN}\left(\text{Conv}_{3\times 3}(\mathbf{F}_{\text{cat}})\right)\right) \in \mathbb{R}^{B\times s\times s\times D}\\ \mathbf{T} &= \text{Flatten}(\mathbf{F}_{\text{emb}}) \in \mathbb{R}^{B\times N\times D}\end{aligned} \tag{1}$$

where B and D denote the input batch size and embedding dimension, respectively.

C. Central-Pixel Guided Dual-Path Multihead Attention Mechanism

Standard self-attention mechanisms compute pairwise correlations between all tokens. This global modeling strategy presents two critical issues when handling HSIC tasks: 1) computational redundancy—edge pixels in neighborhood patches may belong to different classes, and their mutual relationships contribute minimally to central pixel classification and 2) noise introduction—attention connections between irrelevant pixels may introduce interference information. Based on the fundamental principle in HSI classification that "the central pixel determines the patch label," we propose the CP-DMA, which models center-context relationships from both spatial and spectral orthogonal perspectives.

1) Spatial Multihead Attention: The spatial MHA [see Fig. 2(b)] models the relationship between the central pixel and all spatial positions within the neighborhood, capturing spatial contextual information. Given the above token sequence $\mathbf{T}$, we first extract the central position feature as the unique query source

$$\mathbf{t}_c = \mathbf{T}[:, \lfloor N/2 \rfloor, :] \in \mathbb{R}^{B\times 1\times D}. \tag{2}$$

For the MHA mechanism, the embedding dimension D is divided into H heads. For the hth head ($h \in \{1, \ldots, H\}$), the query, key, and value are computed as follows:

$$\begin{aligned}\mathbf{q}_h^{\text{spa}} &= \mathbf{t}_c\mathbf{W}_h^Q \in \mathbb{R}^{B\times 1\times d}\\ \mathbf{K}_h^{\text{spa}} &= \mathbf{T}\mathbf{W}_h^K \in \mathbb{R}^{B\times N\times d}\\ \mathbf{V}_h^{\text{spa}} &= \mathbf{T}\mathbf{W}_h^V \in \mathbb{R}^{B\times N\times d}\end{aligned} \tag{3}$$

where $d = D/H$ denotes the dimension per head, and $\mathbf{W}_h^Q, \mathbf{W}_h^K, \mathbf{W}_h^V \in \mathbb{R}^{D\times d}$ are learnable projection matrices. The basic attention score $\mathcal{A}_h^{\text{spa}}$ is computed via matrix multiplication ($\otimes$), followed by a transpose operation:

$$\mathcal{A}_h^{\text{spa}} = \left(\frac{\mathbf{q}_h^{\text{spa}} \otimes \left(\mathbf{K}_h^{\text{spa}}\right)^T}{\sqrt{d}}\right)^T \in \mathbb{R}^{B\times N\times 1}. \tag{4}$$

However, (4) is purely based on content-based addressing between embedded features, overlooking a crucial spatial structure prior in HSIs—in HSIs, due to the continuity of land cover distribution and spatial autocorrelation, neighboring pixels closer to the central pixel typically have a higher probability of belonging to the same land cover class, and the contextual information they provide is more relevant and reliable. Conversely, distant pixels are more likely to originate from different land covers or even contain interfering noise. To incorporate this inductive bias, we introduce 2-D Gaussian position encoding $\mathbf{G}$ to enhance spatial locality

$$\mathbf{G}_n = \exp\left(-\frac{\|\mathbf{p}_n - \mathbf{p}_c\|^2}{2\sigma^2}\right) \tag{5}$$

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where $\mathbf{p}_n$ and $\mathbf{p}_c$ denote the 2-D coordinates of the nth position and central position, respectively, and σ controls the width of the Gaussian kernel, with a default value of 1. This position encoding assigns higher initial weights to neighboring pixels while preserving the model's ability to adaptively adjust weight distributions. The final SA weights $\mathcal{A}_h^{\text{spa-}G}$ and output $\mathbf{O}_h^{\text{spa}}$ are

$$\begin{aligned}\mathbf{A}_h^{\text{spa-}G} &= \text{softmax}\left(\mathcal{A}_h^{\text{spa}} + \mathbf{G}\right)\\ \mathbf{O}_h^{\text{spa}} &= \mathbf{A}_h^{\text{spa-}G} \otimes \mathbf{V}_h^{\text{spa}} \in \mathbb{R}^{B\times N\times d}.\end{aligned} \tag{6}$$

2) Spectral Multihead Attention: The spectral MHA [see Fig. 2(c)] models interchannel dependencies along the feature spectral (channel) dimension, capturing global correlations among these abstract spectral features. This path also adopts the central-pixel guided principle.

First, the input token sequence $\mathbf{T} \in \mathbb{R}^{B\times N\times D}$ is reshaped into 4-D image form

$$\mathbf{T}_{4\text{–D}} = \text{Reshape}\left(\mathbf{T}\right) \in \mathbb{R}^{B\times s\times s\times D}. \tag{7}$$

Subsequently, unlike the SA path that directly employs linear projection, we adopt depthwise separable convolution [58] stacked with 1×1 pointwise convolution (PWConv) and 3×3 depthwise convolution (DWConv) to generate $\mathbf{Q}^{\text{spe}}$, $\mathbf{K}^{\text{spe}}$, and $\mathbf{V}^{\text{spe}} \in \mathbb{R}^{B\times D\times s\times s}$. In particular, PWConv performs linear mapping in the channel dimension, while DWConv explicitly models local spatial dependencies within each feature channel, thereby achieving richer feature representations with a lower number of parameters. This process can be formalized as

$$\mathbf{Q}^{\text{spe}}, \mathbf{K}^{\text{spe}}, \mathbf{V}^{\text{spe}} = \text{DWConv}\left(\text{PWConv}\left(\mathbf{T}_{4D}\right)\right) \in \mathbb{R}^{B\times s\times s\times D}. \tag{8}$$

Next, similar to SA, we extract the spectral query vector $\mathbf{q}_c^{\text{spe}}$ from the central position of $\mathbf{Q}^{\text{spe}}$. Differently, keys are obtained through global average pooling of $\mathbf{K}^{\text{spe}}$

$$\begin{aligned}\mathbf{q}_c^{\text{spe}} &= \mathbf{Q}^{\text{spe}}\left[:, :, \lfloor s/2\rfloor, \lfloor s/2\rfloor\right] \quad \in \mathbb{R}^{B\times D\times 1}\\ \mathbf{k}_g^{\text{spe}} &= \text{GAP}\left(\mathbf{K}^{\text{spe}}\right) \quad \in \mathbb{R}^{B\times D\times 1}.\end{aligned} \tag{9}$$

Subsequently, the channel dimension of $\mathbf{q}_c^{\text{spe}}$, $\mathbf{k}_g^{\text{spe}}$, and $\mathbf{V}^{\text{spe}}$ is split into H heads, maintaining the same per-head embedding dimension as in SA. For the hth head, we denote the resulting query, key, and value as $\mathbf{q}_h^{\text{spe}}$, $\mathbf{k}_h^{\text{spe}}$, and $\mathbf{V}_h^{\text{spe}} \in \mathbb{R}^{B\times d\times N}$, respectively, and compute the spectral attention $\mathcal{A}_h^{\text{spe}}$ as follows:

$$\mathcal{A}_h^{\text{spe}} = \mathbf{q}_h^{\text{spe}}\left(\mathbf{k}_h^{\text{spe}}\right)^T \in \mathbb{R}^{B\times d\times d}. \tag{10}$$

However, standard self-attention lacks inherent bias for modeling relationships between feature channels. To inject a beneficial inductive bias that encourages the learning of smoother and more structured attention patterns across the channel dimension, we introduce the bidirectional spectral location prior [59] to impose distance-dependent constraints on interchannel correlations

$$\mathbf{S}_{i,j} = \begin{cases}\gamma^{i-j}, & i \geq j\\ \gamma^{j-i}, & i < j\end{cases} \tag{11}$$

where $\mathbf{S}_{i,j}$ denotes the correlation weight between the ith and jth feature channels, i, j are channel indices, and $\gamma \in (0, 1)$ is a decay factor that controls the rate of correlation attenuation with increasing spectral distance. The final spectral attention weights are obtained by

$$\begin{aligned}\mathbf{A}_h^{\text{spe}} &= \text{softmax}\left(\mathcal{A}_h^{\text{spe}} + \mathbf{S}\right)\\ \mathbf{O}_h^{\text{spe}} &= \mathbf{A}_h^{\text{spe}} \otimes \mathbf{V}_h^{\text{spe}} \in \mathbb{R}^{B\times N\times d}.\end{aligned} \tag{12}$$

3) Cross-Dimensional Feature Fusion: The spatial and spectral attention paths capture complementary contextual information from orthogonal perspectives. Effective fusion of these complementary features is crucial for classification performance. We employ the Hadamard product (element-wise multiplication) for feature-level fusion at each head

$$\mathbf{O}_h^{\text{fusion}} = \mathbf{O}_h^{\text{spa}} \odot \mathbf{O}_h^{\text{spe}}. \tag{13}$$

This multiplicative fusion can be interpreted as a gating mechanism, where each path modulates the other by selectively amplifying mutually consistent features while suppressing inconsistent ones, thereby achieving nonlinear second-order interaction. In contrast to linear addition or concatenation, this mechanism enables adaptive cross-dimensional feature calibration with enhanced discriminative capacity. The relevant comparative evaluation and mechanism analysis will be discussed in detail in Section IV-E.

Finally, all heads are concatenated and linearly transformed to obtain the final output

$$\mathbf{F}^{\text{fusion}} = \text{Concat}\left(\mathbf{O}_1^{\text{fusion}}, \ldots, \mathbf{O}_H^{\text{fusion}}\right)\mathbf{W}_{\text{out}} \in \mathbb{R}^{B\times N\times D} \tag{14}$$

where $\mathbf{W}_{\text{out}}$ is the weight matrix for linear transformation.

D. CP-DMA-Based Transformer Encoder

Based on the CP-DMA mechanism, we construct a deep encoder network for hierarchical feature learning. This network consists of L identical encoder blocks stacked sequentially, where each block contains a CP-DMA layer, a position-wise feed-forward network (FFN), and residual connections with layer normalization.

The forward propagation for the lth encoder block ($l \in \{1, 2, \ldots, L\}$) is formulated as

$$\begin{aligned}\mathbf{Z}^l &= \text{LayerNorm}\left(\mathbf{T}^{l-1} + \text{CP-DMA}\left(\mathbf{T}^{l-1}\right)\right)\\ \mathbf{T}^l &= \text{LayerNorm}\left(\mathbf{Z}^l + \text{FFN}\left(\mathbf{Z}^l\right)\right) \in \mathbb{R}^{B\times N\times D}.\end{aligned} \tag{15}$$

The FFN adopts a two-layer architecture with GELU activation to enhance feature transformation

$$\text{FFN}\left(\mathbf{x}\right) = \mathbf{W}_2\left(\text{GELU}\left(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1\right)\right) + \mathbf{b}_2 \in \mathbb{R}^{B\times N\times D} \tag{16}$$

where $\mathbf{W}_1$, $\mathbf{W}_2$, $\mathbf{b}_1$, and $\mathbf{b}_2$ denote the weight matrices and biases of different linear transformations, respectively.

This hierarchical architecture progressively abstracts from local spectral–spatial patterns to global semantic relationships through layer-wise refinement. Final classification is achieved through a classifier consisting of global average pooling and a two-layer MLP

$$\hat{Y} = \text{MLP}\left(\text{GAP}\left(\mathbf{T}^L\right)\right) \tag{17}$$

where $\hat{Y} \in \mathbb{R}^C$ represents the predicted class probabilities.

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E. Feature Fusion Analysis Based on the IB Perspective

To investigate the intrinsic mechanistic differences among different fusion strategies within the CP-DMA module and to elucidate why the multiplication strategy promotes deep feature interaction, this study introduces the information plane analysis method [60] under the framework of information bottleneck (IB) theory [61]. This theory models the feature learning process as an optimization tradeoff problem for the latent representation T: maximizing the target discriminative information $I(Y;T)$ (ensuring classification sufficiency) while minimizing the input redundant information $I(X;T)$ (reducing representation complexity), where $I(\cdot)$ denotes the mutual information (MI). By visualizing the dynamic evolution trajectories of $I(X;T)$ and $I(Y;T)$ during training, the information plane can intuitively depict the model's learning process from "detail fitting" to "core feature compression," thereby quantifying the regulatory efficacy of different fusion strategies on information flow.

Due to the high dimensionality of HSI features and the unknown probability density function (pdf), directly estimating MI is challenging. Therefore, this study employs a matrix-based Rényi α-order entropy ($\alpha = 2$) estimator [62]. This method directly computes entropy and MI via the sample similarity matrix, circumventing the difficulties associated with high-dimensional pdf estimation while maintaining computational efficiency. Given the flattened hidden feature $T \in \mathbb{R}^{n\times d}$, following [61], the specific calculation procedure is elaborated as follows.

First, the Gaussian kernel Gram matrix $K \in \mathbb{R}^{n\times n}$ between samples is constructed, and trace normalization is performed, which is given by

$$K_{ij} = \exp\left(-\frac{\left\|T_i - T_j\right\|^2}{2\sigma^2}\right)$$
$$A = \frac{K}{\mathrm{tr}\,(K)} \tag{18}$$

where $\|T_i - T_j\|^2$ quantifies the feature-space distance of latent representations T_i and T_j. This formulation adopts the same Gaussian kernel form as (5), but it operates on feature distances instead of spatial distances in $\mathbf{G}_n$.

Next, the Rényi second-order entropy and joint entropy are calculated based on matrix spectral theory, defined as follows:

$$H_2(A) = -\log_2 \mathrm{tr}\left(A^2\right)$$
$$H_2(A,B) = H_2\left(\frac{A\odot B}{\mathrm{tr}\,(A\odot B)}\right) \tag{19}$$

where $\odot$ denotes the Hadamard product.

Finally, MI is estimated via the entropy difference, expressed as

$$I(A;B) = H_2(A) + H_2(B) - H_2(A,B) \tag{20}$$

from which $I(X;T)$ and $I(Y;T)$ are computed.

By systematically comparing the evolution trajectories of MI under the three fusion strategies, this study analyzes their dynamic balancing capability of "compressing redundancy while preserving discriminative power" within the information plane, thereby providing a theoretical foundation for understanding the superiority of multiplication fusion.

F. Computational Complexity Analysis

To demonstrate the efficiency of the proposed CP-DMA module, we provide a detailed computational complexity analysis. Let $N = s \times s$ denote the number of spatial tokens, d the input feature dimension, h the number of attention heads, d_h the dimension per head (with $d = hd_h$), and $k = 3$ the kernel size of DWConv. Its overall complexity can be formulated as

$$\begin{aligned}\Omega &= \underbrace{2Ndhd_h + dhd_h}_{\text{Spatial } Q/K/V \text{ proj.}} + \underbrace{2hNd_h}_{\text{Spatial attn}} \\ &\quad + \underbrace{3\left(k^2dN + dhd_hN\right)}_{\text{Spectral } Q/K/V \text{ conv.}} + \underbrace{hd_h^2 + hd_h^2N}_{\text{Spectral attn.}} \\ &\quad + \underbrace{Nd}_{\text{Fusion}} + \underbrace{2d^2N}_{\text{Output proj.}} \\ &= N\left[5\ dhd_h + 2\ hd_h + 3\ k^2d + hd_h^2 + d + 2\ d^2\right] \\ &\quad + dhd_h + hd_h^2. \end{aligned} \tag{21}$$

All dominant terms scale linearly with N, yielding $\Omega = \mathcal{O}(N)$. This is in contrast to standard self-attention ($\mathcal{O}(N^2)$), which computes redundant pairwise correlations across all spatial positions. Relevant experiments are presented in Section IV-F1.

IV. Experiments

A. Dataset Description

1) Indian Pines: The Indian Pines (IP) dataset was acquired by the airborne visible/infrared imaging spectrometer (AVIRIS) in 1992 over a mixed agricultural and forest area in northwestern Indiana, USA. After excluding water absorption and noisy bands, it retains 200 spectral bands spanning 0.4–2.5 μm. The scene comprises 145 $\times$ 145 pixels at 20 m/pixel spatial resolution, with 10 249 labeled samples across 16 crop-related land-cover classes. Its primary challenges stem from severe class imbalance and pronounced intraclass spectral variability.

2) Pavia University: The Pavia University (UP) dataset was captured by the reflective optics system imaging spectrometer (ROSIS) in 2001 over the University of Pavia, Italy. After discarding 12 noisy bands, 103 spectral bands are retained, covering wavelengths from 0.43 to 0.86 μm. The image size is 610 $\times$ 340 pixels with a high spatial resolution of 1.3 m/pixel. The dataset contains 42 776 labeled samples divided into nine urban land-cover classes. It is characterized by regular structures and clear boundaries, making it suitable for evaluating spatial feature extraction capabilities.

3) Houston 2013: The Houston 2013 (HU13) dataset was acquired by the CASI-1500 sensor in 2013, covering the University of Houston campus and surrounding areas, originating from the IEEE GRSS Data Fusion Contest. The data contains 144 bands in the spectral range of 0.38–1.05 μm, with an image size of 340 $\times$ 1905 pixels and a spatial resolution of 2.5 m/pixel. It includes 15 challenging land-cover classes with a total of 16 373 labeled samples. This dataset features high

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TABLE I
NUMBER OF TRAINING AND TESTING SAMPLES FOR IP, UP, HU13, AND HU18 DATASETS

NO.	Indian Pines (IP)			Pavia University (UP)			Houston 2013 (HU13)			Houston 2018 (HU18)		
	Class	Train.	Test.	Class	Train.	Test.	Class	Train.	Test.	Class	Train.	Test.
1	Alfalfa	10	36	Asphalt	10	6621	Healthy grass	10	1241	Healthy grass	50	9749
2	Corn-notill	10	1418	Meadows	10	18639	Stressed grass	10	1244	Stressed grass	50	32452
3	Corn-mintill	10	820	Gravel	10	2089	Synthetic grass	10	687	Synthetic grass	50	634
4	Corn	10	227	Trees	10	3054	Trees	10	1234	Evergreen trees	50	13545
5	Grass-pasture	10	473	Painted metal sheets	10	1335	Soil	10	1232	Deciduous trees	50	4971
6	Grass-trees	10	720	Bare Soil	10	5019	Water	10	315	Soil	50	4466
7	Grass-pasture-m	10	18	Bitumen	10	1320	Residential	10	1258	Water	50	216
8	Hay-windrowed	10	468	Self-Blocking Bricks	10	3672	Commercial	10	1234	Residential buildings	50	39722
9	Oats	10	10	Shadows	10	937	Road	10	1242	Non-residential buildings	50	223702
10	Soybean-notill	10	962				Highway	10	1217	Roads	50	45816
11	Soybean-mintill	10	2445				Railway	10	1225	Sidewalks	50	33979
12	Soybean-clean	10	583				Parking Lot 1	10	1223	Crosswalks	50	1468
13	Wheat	10	195				Parking Lot 2	10	459	Major thoroughfares	50	46298
14	Woods	10	1255				Tennis Court	10	418	Highways	50	9815
15	Buildings-g-t	10	376				Running Track	10	650	Railways	50	6887
16	Stone-Steel-t	10	83							Paved parking lots	50	11450
17										Unpaved parking lots	50	96
18										Cars	50	6497
19										Trains	50	5319
20										Stadium seats	50	6774
	Total	160	10089	Total	90	42686	Total	150	14879	Total	1000	503856

spectral resolution and complex scenes, making it suitable for evaluating model robustness in complex environments.

4) Houston 2018: The Houston 2018 (HU18) dataset, acquired by the National Center for Airborne Laser Mapping (NCALM) over the University of Houston campus and adjacent areas, offers an advanced hyperspectral imaging benchmark. Spanning a spectral range of 0.38–1.05 μm, it includes 50 spectral bands at a ground sampling distance (GSD) of 1 m/pixel. The image size is 601 $\times$ 2384 pixels, containing a total of 504 856 sample points distributed across 20 land-cover categories. Characterized by its high spatial resolution and detailed categorization, HU18 presents a comprehensive resource for evaluating advanced classification algorithms under diverse urban landscapes.

The detailed per-class sample distributions and training/testing partitions for all four datasets are summarized in Table I.

B. Experimental Setup

1) Experimental Environment and Configuration: To ensure a fair comparison, all methods were implemented using the PyTorch framework. Given the large scale of the HU18 dataset, its training and testing were conducted on a server equipped with an Intel[1] Xeon[1] Gold 6459C processor, 94 GB of physical memory, and an NVIDIA RTX A6000 GPU (48-GB VRAM). Conversely, experiments for the IP, UP, and HU13 datasets were executed on a workstation powered by an Intel[1] Core[2] i9-13900HX processor (2.20 GHz), 16 GB of memory, and an NVIDIA GeForce RTX 4060 GPU (8 GB VRAM).

The proposed method utilized the Adam optimizer, with the maximum training epochs uniformly set to 150. Hyperparameters were tailored to the specific characteristics of the datasets: for IP, UP, and HU13, the initial learning rate was set to 1×10^{-3} with a batch size of 32; for HU18, the initial learning rate was adjusted to 5×10^{-4} and the batch size was increased to 64 to balance training stability and efficiency. In all experiments, the learning rate remained constant throughout the training process without any scheduling strategies. No early stopping mechanism was employed, ensuring the models were trained for the full 150 epochs. Furthermore, no data augmentation techniques were introduced during the training phase. For comparison methods, all hyperparameters and implementation details strictly adhered to the optimal configurations reported in their original papers.

2) Evaluation Metrics: The evaluation metrics encompass both classification performance and computational efficiency. Classification performance is measured using overall accuracy (OA), average accuracy (AA), Kappa coefficient ($\mathcal{K}$), and accuracy of each class (AEC) to comprehensively assess overall performance and inter-class balance. Computational efficiency is evaluated through model parameters (Params), floating-point operations (FLOPs) for forward inference, and training/testing time ($T_{\text{train}}/T_{\text{test}}$). All experiments were repeated ten times, with results reported as "mean $\pm$ standard deviation." In the following tables, the best results are shown in **bold**, and the second-best results are underlined.

C. Parameter Analysis

1) Number of PCA Components: Fig. 3(a) illustrates the impact of different PCs on OA across the four datasets, with PCA dimensions ranging from 10 to 150, compared against original features without PCA (noPCA). The results demonstrate that appropriate dimensionality reduction effectively improves classification accuracy. The IP dataset exhibits high sensitivity to spectral information loss; reducing dimensions to 10 yields significantly inferior results compared to the original features. However, performance gradually improves as dimensions increase, reaching a peak at 120 dimensions. Conversely, the UP, HU13, and HU18 datasets exhibit opposite trends, achieving optimal performance with fewer components. In particular, UP and HU13 peak at 30 dimensions, while HU18 achieves its best results at 20 dimensions. Beyond these peaks, accuracy tends to saturate or decline.

[1]Registeredtrademark.
[2]Trademarked

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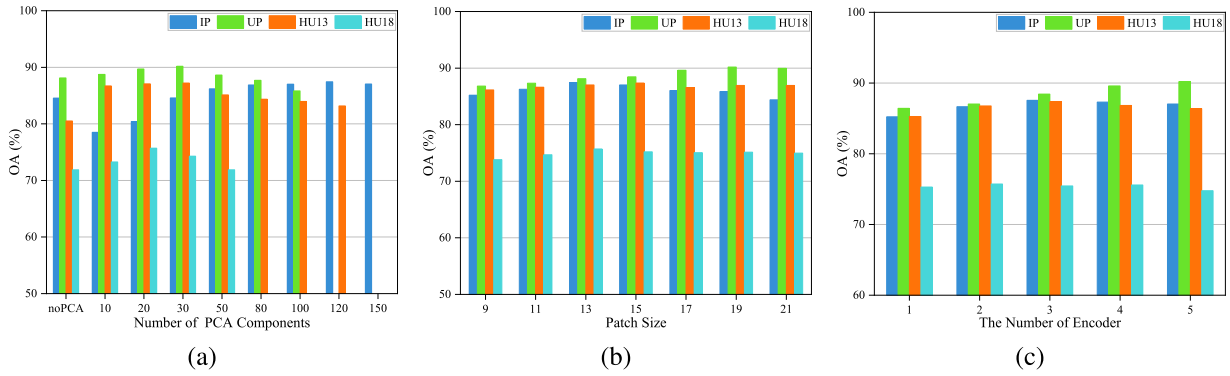


Fig. 3. Impact of different parameters on OA. (a) Number of PCA components. (b) Patch size. (c) Number of encoders.

It is worth noting that for datasets with limited spectral bands (i.e., HU18, UP, and HU13), experimental results for dimensions exceeding their intrinsic spectral resolutions (50, 103, and 144 bands, respectively) are omitted. Based on these observations, the PCA dimensions are set to 120 for IP, 30 for both UP and HU13, and 20 for HU18.

2) Patch Size: Fig. 3(b) investigates the sensitivity of OA to spatial patch size s, ranging from 9×9 to 21×21. It is observed that increasing the patch size initially improves performance by incorporating spatial context, but excessive sizes lead to accuracy degradation due to the inclusion of irrelevant neighbors. In particular, the IP and HU18 datasets achieve peak performance at a relatively small scale ($s = 13$), indicating that their fine-grained or fragmented spatial structures are sensitive to noise introduced by large windows. Conversely, HU13 and UP favor larger receptive fields, reaching optimal accuracy at $s = 15$ and $s = 19$, respectively. This suggests that the latter datasets contain larger homogeneous regions that benefit from broader spatial information. Accordingly, the patch sizes are set to 13 for both IP and HU18, 15 for HU13, and 19 for UP.

3) Number of Encoders: Fig. 3(c) illustrates the impact of network depth, represented by the number of encoders L (ranging from 1 to 5). The results highlight a tradeoff between feature abstraction capability and model complexity. In particular, the HU18 dataset achieves optimal performance with a shallow architecture ($L = 2$), showing rapid saturation where additional layers yield negligible or negative gains. IP and HU13 benefit from moderate depth, peaking at $L = 3$, suggesting that three layers provide sufficient hierarchical representation while avoiding the overfitting associated with deeper structures. In contrast, UP exhibits a continuous upward trend, reaching its best accuracy at $L = 5$, indicating a strong demand for deep semantic features to resolve its complex spatial patterns. Consequently, the number of encoders is set to 3 for both IP and HU13, 5 for UP, and 2 for HU18.

D. Comparison With State-of-the-Art Methods

To validate the effectiveness of the proposed method, this section systematically contrasts its performance against several representative approaches rooted in CNNs, Transformers (encompassing hybrid architectures), GCNs, and the emerging Mamba architecture. The set of compared methods is categorized as follows.

1) CNN-Based Methods:

a) DBDA [63]: This model capitalizes on a dual-branch attention mechanism for spectral–spatial feature extraction. It effectively mitigates the challenge of inadequate feature extraction often encountered in limited-sample scenarios, thus serving as a salient representative of dual-branch networks.

b) CVSSN [25]: This model effectively mines the spectral–spatial information of target pixels through a center vector-guided self-similarity mechanism.

2) Purely Transformer-Based Methods:

a) SpectralFormer (SF) [16]: This model rethinks HSI classification from a sequential perspective, capable of processing both pixel-level and patch-level inputs simultaneously. It is currently one of the most representative algorithms in the Transformer family.

b) miniCAN [24]: As a simplified design of its center attention network (CAN), miniCAN effectively balances computational efficiency and classification accuracy, representing a typical work of the center mechanism.

c) CSIL [26]: Similarly, serving as a representative work of the center mechanism, this model achieves local feature enhancement and global context awareness through the interaction mechanism between center pixels and surrounding pixels.

3) CNN-Transformer Hybrid Architectures:

a) S2FTNet [45]: This model constructs a multibranch spatial–spectral Transformer module, effectively addressing the limitation of a single architecture in capturing both long-range dependencies and local details.

b) HMSSF [44]: This model embeds convolutional operations within the Transformer's self-attention computation and constructs multidimensional attention heads. Through a dynamic feature aggregation mechanism, it achieves the collaborative extraction of hybrid multiscale spatial–spectral features.

c) SQSFormer [57]: This model introduces a rotation-invariant positional encoding module that integrates random central rotation and central relative positional encoding, effectively reducing interference from absolute position and orientation information.

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TABLE II

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE IP DATASET USING TEN LABELED DATA PER CLASS FOR TRAINING

Class	DBDA (2020)	SF (2022)	miniCAN (2022)	CVSSN (2023)	CSIL (2023)	S2FTNet (2023)	HMSSF (2024)	SQSFormer (2024)	LG-GCN (2024)	MambaHSI (2024)	CWSST (2025)	Ours
1	99.72 ± 0.28	91.11 ± 3.42	99.72 ± 0.83	99.72 ± 0.30	99.44 ± 1.67	99.44 ± 0.55	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	99.17 ± 0.82	**100.00 ± 0.00**	**100.00 ± 0.00**
2	79.76 ± 4.15	50.25 ± 5.88	73.65 ± 6.18	78.59 ± 3.45	78.31 ± 5.19	60.94 ± 4.22	77.32 ± 2.15	83.15 ± 1.88	68.51 ± 3.12	69.63 ± 3.05	70.79 ± 4.67	**84.33 ± 2.45**
3	74.95 ± 3.88	53.76 ± 4.12	75.01 ± 9.41	74.72 ± 3.10	72.99 ± 8.18	66.80 ± 4.05	76.59 ± 2.33	**81.49 ± 1.92**	74.32 ± 2.85	72.07 ± 2.88	80.78 ± 3.95	80.96 ± 3.10
4	96.26 ± 1.05	81.67 ± 3.55	95.02 ± 4.70	88.81 ± 2.44	97.22 ± 2.69	95.07 ± 1.15	90.67 ± 1.85	93.39 ± 1.45	91.14 ± 1.25	94.89 ± 1.20	**98.68 ± 0.45**	96.52 ± 0.88
5	88.69 ± 2.12	85.29 ± 2.89	90.49 ± 7.27	89.92 ± 1.88	91.88 ± 4.68	87.99 ± 2.10	84.92 ± 2.55	**91.54 ± 1.35**	88.37 ± 2.05	90.00 ± 1.66	85.26 ± 2.75	88.18 ± 1.25
6	**98.29 ± 0.65**	93.83 ± 1.75	97.75 ± 2.53	97.88 ± 0.85	96.90 ± 2.96	95.14 ± 1.33	97.26 ± 0.95	93.17 ± 1.40	94.60 ± 1.10	95.44 ± 1.15	92.89 ± 1.88	96.68 ± 0.72
7	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	98.89 ± 1.05	**100.00 ± 0.00**	**100.00 ± 0.00**
8	99.74 ± 0.25	86.45 ± 2.66	94.40 ± 5.92	98.06 ± 0.55	94.89 ± 5.46	90.34 ± 1.88	99.79 ± 0.20	**100.00 ± 0.00**	**100.00 ± 0.00**	95.75 ± 1.15	**100.00 ± 0.00**	99.70 ± 0.15
9	99.00 ± 1.00	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
10	81.81 ± 3.45	65.47 ± 4.55	70.99 ± 9.93	72.93 ± 3.15	71.42 ± 8.24	75.14 ± 2.95	72.61 ± 3.20	72.77 ± 3.10	77.74 ± 2.88	75.20 ± 2.90	78.32 ± 3.55	**81.91 ± 2.88**
11	71.94 ± 4.55	50.94 ± 5.12	68.63 ± 8.46	66.58 ± 3.66	68.47 ± 7.94	65.69 ± 4.05	78.41 ± 2.10	76.07 ± 2.25	68.65 ± 3.25	69.25 ± 3.12	76.78 ± 3.75	**80.32 ± 2.65**
12	81.11 ± 3.12	48.44 ± 6.15	72.90 ± 9.84	69.55 ± 3.55	77.72 ± 6.04	60.17 ± 4.55	72.36 ± 2.88	77.02 ± 2.12	77.24 ± 3.45	72.93 ± 2.75	71.58 ± 4.10	**84.94 ± 1.50**
13	99.79 ± 0.21	99.59 ± 0.40	99.44 ± 1.11	92.46 ± 1.88	99.33 ± 0.80	99.90 ± 0.10	96.52 ± 1.15	97.44 ± 0.95	99.69 ± 0.30	99.95 ± 0.05	**100.00 ± 0.00**	99.59 ± 0.25
14	92.74 ± 1.88	88.02 ± 2.45	93.90 ± 6.26	87.04 ± 2.55	94.69 ± 5.39	94.52 ± 1.25	89.73 ± 2.05	**99.36 ± 0.35**	95.04 ± 1.12	92.89 ± 1.55	93.88 ± 1.95	95.99 ± 0.85
15	94.92 ± 1.55	75.11 ± 3.88	93.30 ± 6.55	84.55 ± 2.75	90.32 ± 5.78	85.16 ± 2.66	83.14 ± 2.88	95.20 ± 1.10	92.74 ± 1.65	**97.79 ± 0.65**	94.69 ± 1.45	97.11 ± 0.68
16	99.40 ± 0.60	99.76 ± 0.24	99.76 ± 0.72	98.92 ± 0.55	99.28 ± 0.96	**100.00 ± 0.00**	99.57 ± 0.42	**100.00 ± 0.00**	99.88 ± 0.12	98.67 ± 0.88	**100.00 ± 0.00**	99.40 ± 0.35
OA (%)	83.64 ± 1.99	66.46 ± 3.21	80.33 ± 1.88	80.54 ± 1.62	81.16 ± 2.02	81.79 ± 2.33	81.30 ± 1.91	84.37 ± 2.40	82.66 ± 2.45	80.12 ± 2.54	83.33 ± 2.19	**87.38 ± 2.11**
AA (%)	91.13 ± 1.22	79.36 ± 1.18	89.06 ± 0.85	87.48 ± 1.09	89.55 ± 1.15	86.02 ± 1.14	89.05 ± 0.86	91.29 ± 0.85	88.97 ± 1.24	89.09 ± 0.90	90.23 ± 1.21	**92.85 ± 0.84**
$\mathcal{K} \times 100$	81.54 ± 2.19	62.45 ± 3.44	77.76 ± 2.07	78.31 ± 2.30	78.71 ± 2.24	79.20 ± 2.51	78.88 ± 2.06	82.27 ± 2.61	80.39 ± 2.37	77.65 ± 2.74	81.20 ± 2.40	**85.73 ± 2.35**

TABLE III

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE UP DATASET USING TEN LABELED DATA PER CLASS FOR TRAINING

Class	DBDA (2020)	SF (2022)	miniCAN (2022)	CVSSN (2023)	CSIL (2023)	S2FTNet (2023)	HMSSF (2024)	SQSFormer (2024)	LG-GCN (2024)	MambaHSI (2024)	CWSST (2025)	Ours
1	**92.95 ± 2.15**	55.07 ± 7.85	64.00 ± 8.10	78.89 ± 3.55	72.30 ± 8.46	78.24 ± 4.05	83.77 ± 3.10	71.31 ± 5.15	72.80 ± 4.22	84.57 ± 2.65	53.20 ± 6.12	83.35 ± 2.88
2	81.88 ± 3.55	72.16 ± 5.12	81.41 ± 7.07	84.38 ± 2.88	83.71 ± 5.72	86.01 ± 2.05	84.85 ± 2.35	85.89 ± 2.12	84.57 ± 2.10	87.66 ± 1.88	76.50 ± 4.25	**89.70 ± 1.65**
3	84.05 ± 3.12	76.73 ± 4.45	77.30 ± 5.93	83.12 ± 2.95	78.88 ± 7.37	83.38 ± 2.88	85.61 ± 2.40	80.23 ± 3.55	82.95 ± 2.75	80.53 ± 3.10	75.54 ± 4.55	**87.98 ± 1.95**
4	93.77 ± 1.88	88.44 ± 3.05	91.40 ± 4.36	**96.61 ± 1.12**	90.66 ± 4.33	89.58 ± 2.15	90.94 ± 1.55	83.36 ± 3.12	87.32 ± 1.65	80.38 ± 3.45	85.76 ± 2.88	89.24 ± 1.85
5	98.19 ± 0.65	**99.99 ± 0.01**	**99.99 ± 0.04**	98.28 ± 0.55	99.98 ± 0.05	99.73 ± 0.15	99.96 ± 0.04	99.85 ± 0.12	99.85 ± 0.10	99.80 ± 0.15	99.85 ± 0.10	99.98 ± 0.02
6	**98.63 ± 0.55**	65.00 ± 6.78	82.70 ± 6.44	80.48 ± 3.65	88.34 ± 6.94	88.85 ± 2.25	87.10 ± 2.55	90.83 ± 1.88	89.31 ± 2.12	**96.23 ± 1.10**	88.40 ± 2.95	97.16 ± 0.85
7	98.73 ± 0.45	89.71 ± 2.15	97.02 ± 2.31	95.49 ± 1.15	98.64 ± 1.32	**99.59 ± 0.20**	99.56 ± 0.22	98.02 ± 0.65	99.42 ± 0.25	90.45 ± 1.95	95.91 ± 1.10	99.43 ± 0.25
8	74.40 ± 5.12	38.64 ± 8.95	58.28 ± 8.93	79.20 ± 3.88	65.13 ± 9.76	72.56 ± 4.88	77.84 ± 3.55	72.25 ± 4.65	75.10 ± 4.55	**88.96 ± 2.15**	65.13 ± 5.88	88.67 ± 2.45
9	99.12 ± 0.45	93.08 ± 1.88	**99.59 ± 0.22**	99.58 ± 0.25	98.87 ± 0.33	98.80 ± 0.55	99.25 ± 0.40	93.20 ± 1.45	97.78 ± 0.65	98.48 ± 0.55	93.92 ± 1.25	98.27 ± 0.50
OA (%)	88.41 ± 3.91	69.94 ± 3.06	78.85 ± 3.06	84.55 ± 3.57	82.45 ± 2.90	84.74 ± 2.36	85.46 ± 2.49	82.93 ± 3.94	84.26 ± 2.74	88.13 ± 2.19	78.33 ± 2.81	**90.20 ± 3.55**
AA (%)	91.30 ± 1.78	75.42 ± 3.78	83.52 ± 1.15	88.45 ± 2.41	86.28 ± 1.36	88.53 ± 1.40	89.88 ± 1.38	86.10 ± 3.03	87.68 ± 1.47	89.67 ± 1.89	81.57 ± 2.15	**92.64 ± 1.91**
$\mathcal{K} \times 100$	85.14 ± 4.81	67.17 ± 3.62	72.95 ± 3.50	80.05 ± 4.35	77.47 ± 3.46	80.25 ± 2.61	81.37 ± 3.03	77.85 ± 4.95	79.84 ± 3.36	84.62 ± 2.71	72.28 ± 3.35	**87.30 ± 4.14**

TABLE IV

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE HU13 DATASET USING TEN LABELED DATA PER CLASS FOR TRAINING

Class	DBDA (2020)	SF (2022)	miniCAN (2022)	CVSSN (2023)	CSIL (2023)	S2FTNet (2023)	HMSSF (2024)	SQSFormer (2024)	LG-GCN (2024)	MambaHSI (2024)	CWSST (2025)	Ours
1	91.02 ± 2.15	81.85 ± 4.55	88.10 ± 4.40	92.18 ± 2.10	88.97 ± 5.30	84.05 ± 3.66	86.70 ± 3.05	**94.68 ± 1.45**	87.63 ± 3.12	89.36 ± 2.55	92.91 ± 2.88	89.89 ± 1.85
2	87.86 ± 2.88	91.48 ± 2.55	90.51 ± 4.66	89.20 ± 3.10	94.52 ± 5.52	88.92 ± 2.95	91.32 ± 2.45	97.03 ± 0.95	93.43 ± 1.88	**97.43 ± 0.88**	93.41 ± 2.12	94.25 ± 0.65
3	99.24 ± 0.35	98.69 ± 0.65	91.70 ± 10.45	98.14 ± 0.85	98.72 ± 2.19	97.38 ± 0.95	95.05 ± 1.55	99.56 ± 0.25	98.65 ± 0.55	98.31 ± 0.75	95.49 ± 1.25	**99.90 ± 0.05**
4	**96.56 ± 1.05**	87.57 ± 3.55	93.69 ± 3.37	87.80 ± 3.25	93.34 ± 3.44	90.44 ± 2.45	94.07 ± 1.65	85.49 ± 3.88	92.15 ± 2.15	88.52 ± 2.88	90.28 ± 2.55	90.64 ± 2.35
5	98.88 ± 0.45	94.66 ± 1.65	98.12 ± 3.98	97.59 ± 0.85	99.51 ± 1.06	98.90 ± 0.55	99.09 ± 0.45	99.68 ± 0.15	99.61 ± 0.20	99.76 ± 0.12	**100.00 ± 0.00**	99.70 ± 0.10
6	89.33 ± 2.55	89.52 ± 2.88	87.81 ± 5.33	88.44 ± 2.95	92.92 ± 5.12	90.60 ± 2.40	88.51 ± 2.85	**93.97 ± 1.65**	90.89 ± 2.15	86.19 ± 3.12	85.48 ± 3.55	88.76 ± 1.75
7	85.12 ± 3.12	61.21 ± 6.85	59.44 ± 9.44	77.83 ± 4.55	77.94 ± 10.20	77.50 ± 4.12	**86.38 ± 2.95**	76.68 ± 4.45	82.72 ± 4.25	78.54 ± 4.10	61.61 ± 6.12	83.15 ± 1.55
8	58.57 ± 6.55	51.80 ± 8.12	62.97 ± 6.47	49.27 ± 8.45	66.49 ± 6.46	56.97 ± 6.15	56.90 ± 6.55	**68.15 ± 4.12**	62.83 ± 5.88	57.86 ± 6.45	64.18 ± 7.55	63.74 ± 1.88
9	**85.31 ± 3.05**	38.94 ± 9.45	66.98 ± 7.85	79.04 ± 4.10	69.23 ± 16.42	67.97 ± 5.55	67.57 ± 5.65	74.91 ± 4.25	67.02 ± 5.12	76.97 ± 4.05	63.77 ± 7.12	77.83 ± 1.95
10	86.89 ± 2.95	46.69 ± 8.55	71.76 ± 6.59	81.26 ± 3.45	75.18 ± 8.47	82.22 ± 3.12	79.61 ± 3.66	71.24 ± 4.88	80.12 ± 3.88	83.52 ± 2.95	63.43 ± 7.25	**88.42 ± 1.10**
11	79.77 ± 3.66	46.89 ± 8.88	69.17 ± 12.44	71.85 ± 4.55	82.62 ± 13.22	77.86 ± 3.88	76.24 ± 3.95	77.43 ± 3.55	83.55 ± 3.10	82.69 ± 3.12	59.02 ± 7.88	**86.53 ± 1.25**
12	78.12 ± 4.10	53.72 ± 7.15	74.28 ± 4.67	**87.03 ± 2.65**	66.17 ± 10.52	83.60 ± 3.12	65.82 ± 5.12	63.97 ± 5.55	74.35 ± 4.55	76.78 ± 3.88	74.49 ± 4.65	79.02 ± 2.66
13	88.54 ± 2.45	71.24 ± 5.12	91.96 ± 1.71	94.75 ± 1.65	91.00 ± 2.45	91.68 ± 2.10	95.16 ± 1.35	93.03 ± 1.88	**96.14 ± 1.25**	89.96 ± 2.45	89.11 ± 2.75	89.60 ± 1.54
14	**100.00 ± 0.00**	93.01 ± 2.12	99.40 ± 0.98	99.52 ± 0.25	99.88 ± 0.29	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	99.86 ± 0.10	**100.00 ± 0.00**	**100.00 ± 0.00**	99.95 ± 0.04
15	98.49 ± 0.55	99.91 ± 0.05	99.55 ± 0.64	99.48 ± 0.20	99.98 ± 0.05	99.82 ± 0.10	99.23 ± 0.35	**100.00 ± 0.00**	99.91 ± 0.05	99.38 ± 0.30	**100.00 ± 0.00**	**100.00 ± 0.00**
OA (%)	86.72 ± 1.85	70.02 ± 2.48	81.39 ± 1.86	83.96 ± 2.64	84.08 ± 2.43	83.50 ± 2.14	83.10 ± 1.58	86.08 ± 2.06	85.46 ± 2.55	84.87 ± 1.92	79.11 ± 2.24	**87.38 ± 1.16**
AA (%)	88.25 ± 1.38	73.81 ± 2.07	83.03 ± 1.20	86.23 ± 2.15	86.43 ± 1.95	85.86 ± 0.63	85.44 ± 1.36	86.39 ± 1.67	87.26 ± 1.43	87.02 ± 1.57	82.21 ± 1.83	**88.76 ± 0.65**
$\mathcal{K} \times 100$	85.65 ± 2.00	67.59 ± 2.68	79.82 ± 1.41	82.66 ± 2.85	82.80 ± 2.62	82.18 ± 1.08	81.72 ± 1.70	84.95 ± 2.23	84.19 ± 2.48	83.51 ± 1.89	78.35 ± 2.41	**86.37 ± 1.25**

d) CWSST [47]: This model designs a cross-window attention mechanism based on grouped tokens, enabling information to be aggregated and propagated across different windows, thereby acquiring more refined features.

4) GCN-Based Methods:

a) LG-GCN [42]: This model builds a dual-branch GCN at both the superpixel-level and pixel-level, alleviating the difficulty for a single graph structure to simultaneously account for local details and global semantics.

5) Mamba-Based Methods:

a) MambaHSI [55]: As the first exploration of the Mamba architecture in the hyperspectral domain, this model demonstrates the potential for linear-complexity classification by leveraging the sequential modeling capabilities of SSMs.

1) Quantitative Performance Comparison: Tables II–V present the per-class accuracy, OA, AA, and $\mathcal{K}$ of the proposed method and all competing methods on the four datasets.

On the IP dataset, the experiments were conducted using 10 samples per class for training. Traditional CNN-based methods (e.g., DBDA and CVSSN) outperformed the pure Transformer-based SF, indicating that convolutional operations still possess an irreplaceable advantage in effectively modeling local spatial features. Among them, CVSSN performed slightly worse than DBDA, primarily because it relies mainly on the local receptive field of convolutional kernels, whereas DBDA mitigates this limitation by incorporating a self-attention mechanism to fuse long-range contextual information. Subsequently, CNN-Transformer hybrid methods (e.g., S2FTNet, HMSSF, and SQSFormer) attempted to balance local and global features. SQSFormer achieved leading performance in most classes (e.g., Class 5-"Grass-pasture," Class 8-"Hay-windrowed," and Class 14-"Woods") and became the previous state-of-the-art method with an OA of 84.37%. However, its discriminative power for some challenging land covers (e.g., Class 4-"Corn, " Class 10-"Soybean-notill," and Class 11-"Soybean-mintill") remained insufficient. Recent center-focused attention methods (miniCAN and CSIL) demonstrated moderate improvements,

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TABLE V

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE HU18 DATASET USING 50 LABELED DATA PER CLASS FOR TRAINING

Class	DBDA (2020)	SF (2022)	miniCAN (2022)	CVSSN (2023)	CSIL (2023)	S2FTNet (2023)	HMSSF (2024)	SQSFormer (2024)	LG-GCN (2024)	MambaHSI (2024)	CWSST (2025)	Ours
1	**68.27 ± 5.05**	62.43 ± 6.04	63.86 ± 7.57	63.04 ± 6.72	65.42 ± 8.54	56.68 ± 9.34	50.62 ± 6.39	48.81 ± 4.13	52.88 ± 3.47	56.00 ± 5.38	54.12 ± 6.21	57.17 ± 6.46
2	**90.12 ± 2.80**	81.37 ± 4.92	76.08 ± 4.28	87.03 ± 4.28	77.26 ± 6.78	80.41 ± 11.36	87.63 ± 3.77	85.74 ± 2.48	86.82 ± 3.36	87.64 ± 1.79	85.10 ± 5.44	88.00 ± 4.97
3	**96.73 ± 3.24**	43.89 ± 18.27	79.95 ± 6.07	86.83 ± 11.55	86.94 ± 4.08	70.87 ± 21.24	76.80 ± 29.13	75.75 ± 13.16	79.58 ± 22.01	61.86 ± 21.34	77.55 ± 13.88	85.68 ± 25.33
4	**83.28 ± 2.74**	79.67 ± 6.62	80.76 ± 3.14	80.90 ± 3.19	82.39 ± 3.63	62.84 ± 17.11	71.67 ± 6.16	70.63 ± 8.44	73.02 ± 6.41	78.21 ± 7.03	75.30 ± 9.66	75.72 ± 3.53
5	49.62 ± 8.32	42.22 ± 5.66	45.74 ± 9.52	39.61 ± 9.70	48.95 ± 8.29	34.85 ± 6.36	41.12 ± 11.43	38.84 ± 9.04	43.73 ± 6.97	39.94 ± 6.89	40.88 ± 7.15	**50.49 ± 5.78**
6	73.55 ± 12.80	25.73 ± 4.59	59.02 ± 17.82	58.13 ± 13.47	71.31 ± 14.77	75.09 ± 23.84	70.66 ± 14.51	72.47 ± 17.15	76.78 ± 3.91	76.49 ± 15.08	71.20 ± 14.66	**79.20 ± 18.97**
7	83.21 ± 18.54	15.05 ± 8.23	75.95 ± 16.14	61.14 ± 18.68	79.20 ± 17.52	36.09 ± 11.95	49.56 ± 28.36	44.73 ± 26.76	67.26 ± 19.91	69.07 ± 17.92	65.15 ± 20.14	**87.02 ± 16.85**
8	**69.20 ± 4.75**	62.15 ± 5.01	61.93 ± 5.70	68.98 ± 5.03	65.11 ± 5.06	61.10 ± 5.33	67.25 ± 5.98	64.07 ± 6.17	67.94 ± 6.63	66.22 ± 7.03	68.55 ± 6.88	67.26 ± 5.80
9	97.62 ± 0.62	93.99 ± 1.03	87.01 ± 4.52	**97.72 ± 0.37**	92.26 ± 1.80	93.86 ± 2.10	96.75 ± 1.09	97.06 ± 0.60	97.27 ± 0.83	97.16 ± 0.53	97.10 ± 0.71	97.45 ± 0.77
10	66.04 ± 3.29	45.79 ± 2.75	37.05 ± 6.46	57.84 ± 3.13	40.54 ± 6.65	47.74 ± 8.78	57.73 ± 7.76	56.80 ± 4.70	**66.35 ± 5.69**	55.87 ± 7.32	58.66 ± 6.55	58.60 ± 8.20
11	48.52 ± 4.12	43.11 ± 3.41	45.91 ± 5.58	47.04 ± 4.36	47.65 ± 4.30	29.77 ± 8.35	39.47 ± 5.01	44.93 ± 4.13	48.43 ± 4.61	43.61 ± 5.22	45.20 ± 5.88	**49.85 ± 7.49**
12	**6.88 ± 2.10**	3.64 ± 0.66	4.40 ± 2.45	5.87 ± 0.76	5.69 ± 3.14	5.33 ± 1.69	6.18 ± 0.85	5.40 ± 1.25	**6.88 ± 1.05**	5.72 ± 1.27	6.15 ± 1.33	6.56 ± 1.25
13	73.81 ± 3.10	52.88 ± 2.08	62.94 ± 3.71	71.77 ± 4.81	68.95 ± 3.02	63.35 ± 1.80	75.49 ± 3.51	69.33 ± 4.29	**77.86 ± 2.88**	67.98 ± 2.29	72.55 ± 3.66	70.00 ± 3.05
14	58.13 ± 8.06	42.24 ± 9.26	46.91 ± 9.43	**66.37 ± 8.74**	57.55 ± 8.68	59.88 ± 8.29	57.08 ± 6.87	56.26 ± 6.44	58.38 ± 7.72	58.68 ± 5.11	60.14 ± 7.55	60.82 ± 7.54
15	88.50 ± 5.96	43.10 ± 3.61	68.36 ± 6.84	52.82 ± 8.65	73.13 ± 7.59	84.23 ± 6.50	79.22 ± 13.13	80.27 ± 7.32	88.01 ± 6.63	85.06 ± 4.68	83.55 ± 6.44	**91.64 ± 4.61**
16	**81.39 ± 5.33**	46.91 ± 4.22	56.06 ± 5.46	65.55 ± 7.70	70.80 ± 3.04	65.33 ± 23.02	79.66 ± 7.43	79.64 ± 7.57	79.98 ± 5.61	77.64 ± 7.53	78.66 ± 8.55	78.91 ± 7.67
17	**26.14 ± 19.82**	2.93 ± 1.97	8.46 ± 11.72	12.42 ± 7.67	13.20 ± 19.17	15.11 ± 13.33	15.57 ± 11.97	16.96 ± 12.60	22.52 ± 19.07	12.69 ± 11.52	18.88 ± 14.22	14.57 ± 8.91
18	54.99 ± 10.73	34.52 ± 4.25	50.24 ± 10.11	56.08 ± 9.61	53.13 ± 9.35	49.87 ± 10.64	45.92 ± 10.24	50.20 ± 13.78	47.71 ± 10.35	**57.47 ± 11.86**	52.66 ± 9.55	53.52 ± 10.79
19	68.55 ± 16.02	38.03 ± 8.99	58.30 ± 14.25	**88.25 ± 7.97**	60.44 ± 10.64	70.80 ± 11.36	63.43 ± 19.05	75.00 ± 7.08	68.92 ± 11.56	79.38 ± 11.04	74.22 ± 10.88	86.00 ± 7.09
20	69.56 ± 14.24	40.97 ± 6.81	57.19 ± 13.50	63.38 ± 10.59	61.50 ± 13.38	**73.48 ± 13.53**	62.07 ± 8.27	61.76 ± 10.40	68.97 ± 6.48	62.90 ± 8.60	67.55 ± 9.14	70.51 ± 10.89
OA (%)	**76.12 ± 1.98**	56.80 ± 2.22	66.35 ± 2.18	72.59 ± 2.30	70.23 ± 1.73	64.74 ± 5.24	71.06 ± 2.61	72.24 ± 1.87	73.43 ± 1.97	72.64 ± 2.09	70.02 ± 2.48	75.72 ± 2.10
AA (%)	**67.71 ± 2.39**	44.03 ± 1.66	57.06 ± 2.19	61.54 ± 2.92	61.17 ± 1.62	56.33 ± 2.80	59.69 ± 2.90	59.73 ± 3.55	63.97 ± 2.45	61.98 ± 3.34	58.76 ± 2.07	66.45 ± 2.49
$\mathcal{K} \times 100$	**70.55 ± 2.21**	50.12 ± 2.08	59.45 ± 2.17	66.73 ± 2.41	63.69 ± 1.79	57.46 ± 5.65	64.75 ± 2.87	65.80 ± 1.95	67.58 ± 2.09	66.34 ± 2.33	63.51 ± 2.68	69.82 ± 2.35

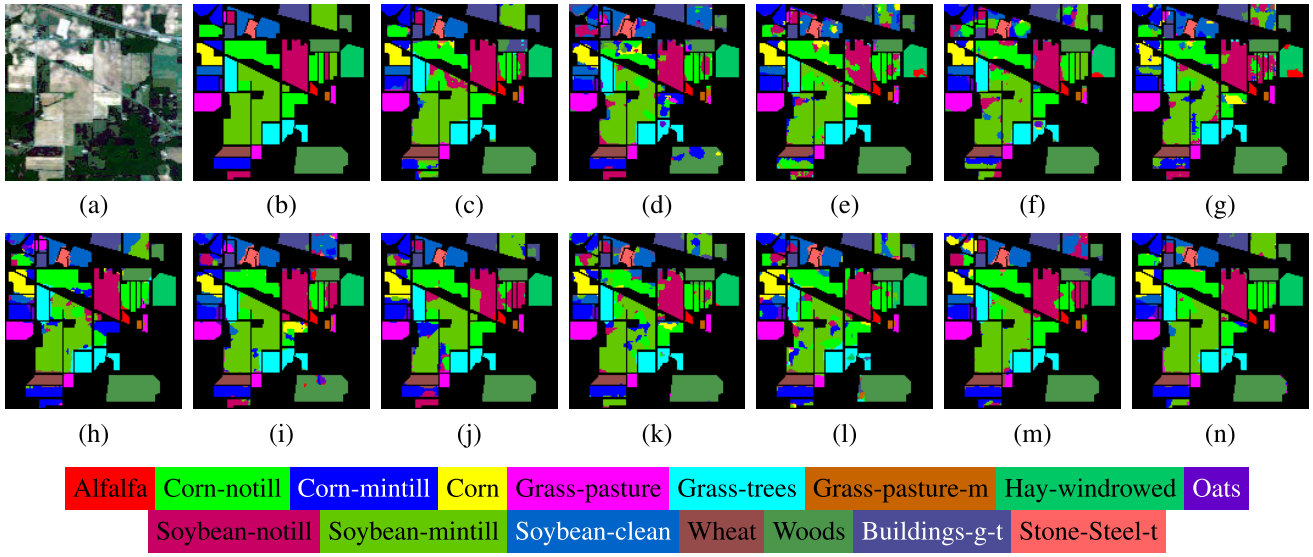


Fig. 4. Visual classification results of IP with compared methods. (a) False-color composite image. (b) Ground-truth map. (c) DBDA. (d) SF. (e) miniCAN. (f) CVSSN. (g) CSIL. (h) S2FTNet. (i) HMSSF. (j) SQSFormer. (k) LG-GCN. (l) MambaHSI. (m) CWSST. (n) Ours.

achieving OAs of 80.33% and 81.16%, respectively, yet both underperformed compared to SQSFormer and the proposed method. miniCAN, which emphasizes central pixel spectral features and similar neighborhood associations through scaled dot-product central attention, struggled particularly with challenging agricultural classes (e.g., Class 2-"Corn-notill" and Class 11-"Soybean-mintill"), suggesting that its similarity-based neighborhood selection may be insufficient when dealing with spectrally ambiguous crops. CSIL, employing hierarchical region sampling to model center-to-surrounding interactions, achieved slightly better results but still fell short in capturing fine-grained discriminative features for complex agricultural scenarios. Recent methods based on graph structures and SSMs (e.g., LG-GCN and MambaHSI) demonstrated new potential. The former achieved an OA of 82.66%, and the latter attained a high accuracy of 97.79% in Class 15 ("Buildings-g-t"). However, both showed significant weaknesses, with accuracies below 80% in classes such as Class 2 ("Corn-notill") and Class 11 ("Soybean-mintill"). The core reasons may lie in: the graph structure of LG-GCN being prone to distortion due to sparse labels, and the sequential modeling path of MambaHSI struggling to capture complex spatial relationships, leading to insufficient adaptability to feature distributions. In contrast, the proposed method achieved the highest accuracy in 7 out of 16 classes and attained overall metrics of OA 87.38%, AA 92.85%, and $\mathcal{K}$ 85.73%, surpassing all existing methods. This verifies the superiority of the proposed method in balancing local details and global consistency.

On the UP and HU13 datasets, also trained with 10 samples per class, the proposed method again demonstrated advantages, outperforming all compared methods with OAs of 90.20% and 87.38%, respectively. Although it achieved the highest accuracy in only 2 of UP's 9 classes and 4 of HU13's 15 classes, its strength lies in the balanced overall performance and effective recognition of challenging land covers (e.g., Classes 2 and 3 in UP and Classes 10 and 11 in HU13). It is noteworthy that the CNN-based DBDA achieved suboptimal results on both datasets, surpassing most emerging comparative methods. This benefits from the powerful synergy of its dense connections and self-attention mechanism, which enables a skillful balance between local feature learning and

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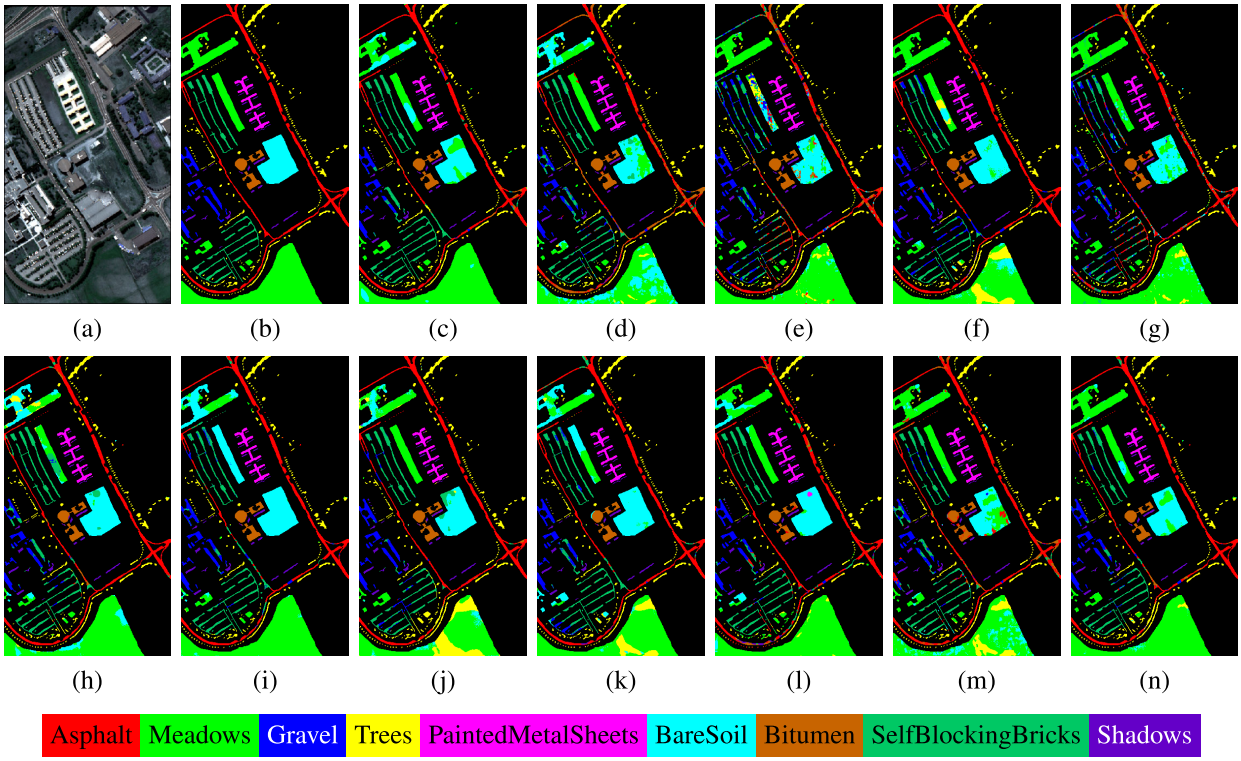


Fig. 5. Visual classification results of UP with compared methods. (a) False-color composite image. (b) Ground-truth map. (c) DBDA. (d) SF. (e) miniCAN. (f) CVSSN. (g) CSIL. (h) S2FTNet. (i) HMSSF. (j) SQSFormer. (k) LG-GCN. (l) MambaHSI. (m) CWSST. (n) Ours.

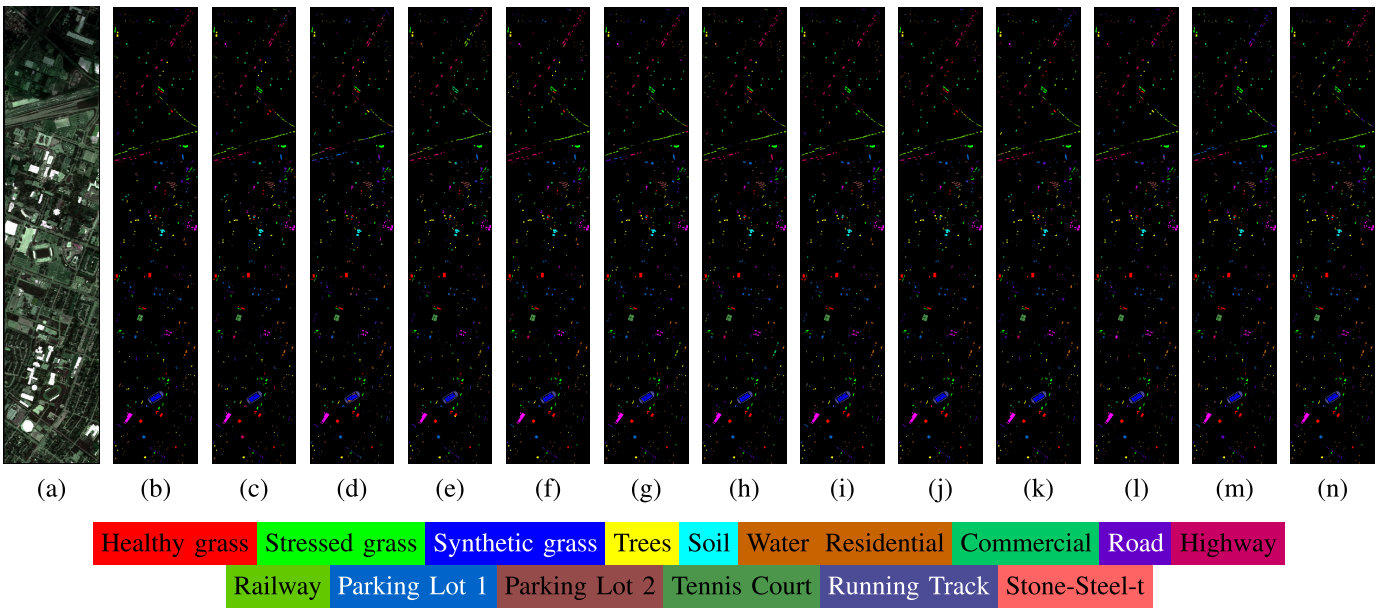


Fig. 6. Visual classification results of HU13 with compared methods. (a) False-color composite image. (b) Ground-truth map. (c) DBDA. (d) SF. (e) miniCAN. (f) CVSSN. (g) CSIL. (h) S2FTNet. (i) HMSSF. (j) SQSFormer. (k) LG-GCN. (l) MambaHSI. (m) CWSST. (n) Ours.

long-range context integration even under limited samples. However, when shifting to the HU18 dataset, which has more abundant samples (50 per class) and more complex categories, the proposed method ranked second with an OA of 75.72%, slightly lower than DBDA (76.12%). This slight suboptimal performance hints at a potential limitation: when training data is relatively sufficient, the inherent advantage of our attention mechanism in sparse data settings might be offset by the strong generalization capability of deep CNNs (e.g., DBDA) on high-quality, high-resolution urban data. Nevertheless, the proposed method achieved the best performance on three out of the four tested datasets and demonstrated near-optimal performance on HU18, validating its effectiveness and robustness across diverse scenarios.

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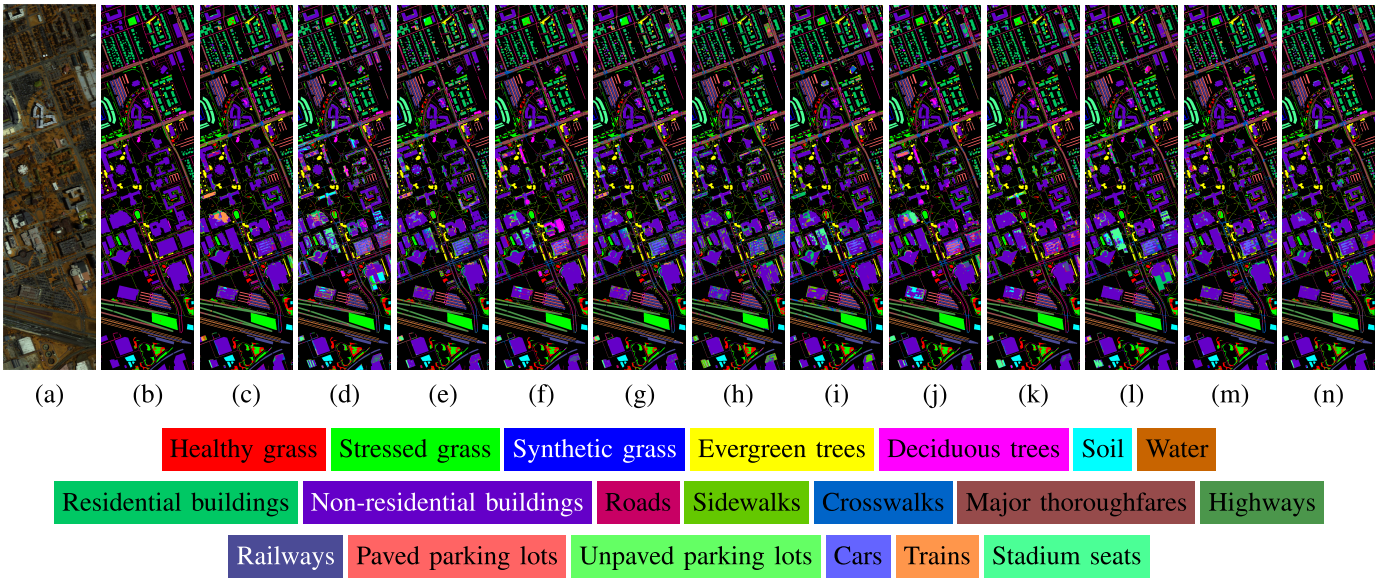


Fig. 7. Visual classification results of HU18 with compared methods. (a) False-color composite image. (b) Ground-truth map. (c) DBDA. (d) SF. (e) miniCAN. (f) CVSSN. (g) CSIL. (h) S2FTNet. (i) HMSSF. (j) SQSFormer. (k) LG-GCN. (l) MambaHSI. (m) CWSST. (n) Ours.

TABLE VI
COMPARISONS ON MODEL EFFICIENCY OF DIFFERENT METHODS ON IP DATASETS

Metric	DBDA	SF	miniCAN	CVSSN	CSIL	S2FTNet	HMSSF	SQSFormer	LG-GCN	MambaHSI	CWSST	ours
T_{train} (s)	74.37	109.39	126.77	**4.26**	143.51	23.54	35.68	10.59	76.53	124.73	9.69	11.17
T_{test} (ms)	4.99	9.74	6.21	3.48	7.05	5.89	75.90	2.07	6.04	69	5.12	4.72
Params (MB)	0.38	0.60	0.44	0.26	0.76	0.75	1.28	**0.23**	0.52	0.41	1.21	0.28
FLOPs (GB)	1.76	4.61	1.53	**0.68**	2.85	1.47	2.41	2.77	5.26	4.49	12.89	1.41

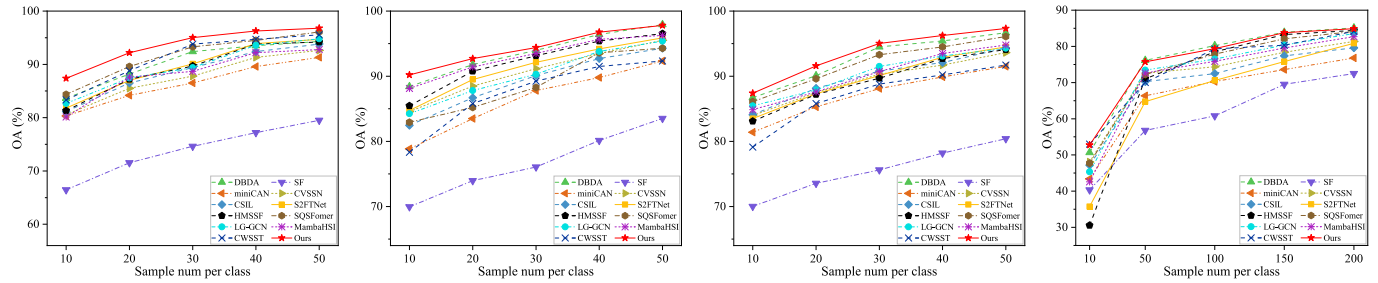


Fig. 8. Impact of different numbers of training samples on OA. (a) IP dataset. (b) UP dataset. (c) HU13 dataset. (d) HU18 dataset.

2) Qualitative Comparison: Figs. 4–7 visualizes the classification maps, three-band false-color composites, and ground truth maps for all methods on IP, UP, HU13, and HU18 datasets. Overall, the visualization results across different methods and datasets are consistent with the quantitative results in the Tables II–V, with our method generating smoother and clearer classification maps. In particular, examining the IP dataset, by comparing regions containing "Corn-notill," "Grass-trees," and "Soybean-mintill," the proposed method demonstrates smoother and clearer performance in these regions compared to the second-best method and others.

3) Efficiency Comparison: As shown in Table VI, we compare different methods' performance on the IP dataset in terms of runtime, model complexity, and computational cost. CVSSN, SQSFormer, and the proposed method exhibit lower computational complexity and superior runtime efficiency, with our method achieving comparable efficiency while providing superior classification performance. Notably, MambaHSI's inference time is highlighted to emphasize its shortest inference duration. As described, MambaHSI operates as an image-level classification task, processing entire images directly during inference, while other methods (including ours) compute based on individual batches, resulting in longer inference times. Except for this metric, all other performance indicators were compared under unified standards.

4) Robustness Comparison: To validate the robustness of the proposed method, we conducted comparative experiments

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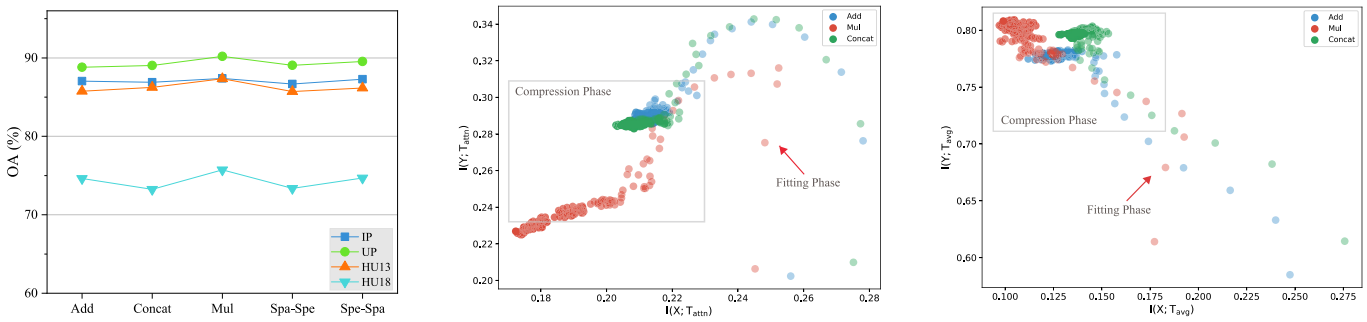


Fig. 9. Comparative analysis of different feature fusion strategies via classification performance and information plane. (a) OA comparison across all datasets. (b) Information plane of feature T_{attn} on the IP dataset. (c) Information plane of feature T_{avg} on the IP dataset.

with varying training sample sizes across four datasets. As shown in Fig. 8, the overall trend indicates that the performance of all methods improves with an increase in training samples. In particular, in the IP, UP, and HU13 datasets, when the number of training samples per class increases from 10 to 50, our method demonstrates particularly significant advantages under limited sample conditions (10–20 samples per class). Even when the sample size reaches 50, our method maintains its leading position, verifying its excellent stability and scalability. However, in the HU18 dataset, when the training sample size is expanded to the range of 10–200, DBDA exhibits a slight superiority over our method in the high-sample interval (≥50). A plausible explanation is that HU18 features high category granularity and large intraclass variation; with relatively sufficient samples, DBDA's 3-D dense convolution and self-attention structures can more effectively model local details, thereby gaining a minor advantage. Nevertheless, our method maintains highly stable performance throughout the entire sample range, further confirming its practical value and robustness in annotation-limited scenarios.

E. Comparison of Feature Fusion Strategies

To comprehensively evaluate the effectiveness of the element-wise multiplication fusion strategy within the proposed CP-DMA module, this section provides a systematic analysis from two complementary perspectives: performance comparison and mechanism visualization.

1) Performance Comparison: A quantitative comparison of the OA was conducted for five fusion strategies across four datasets: element-wise multiplication (Mul), addition (Add), concatenation (Concat), and two sequential attention pathways (Spa–Spe and Spe–Spa). As illustrated in Fig. 9(a), Mul consistently achieved the best classification performance on all datasets, with its accuracy uniformly surpassing that of the two linear fusion strategies (Add and Concat). This indicates that, within the dual-path attention framework, the second-order nonlinear interaction introduced by multiplication is more effective than linear fusion. Furthermore, its superiority over the sequentially dependent processing methods also validates the design efficacy of the parallel fusion structure, which leverages multiplication for instantaneous cross-path calibration.

2) Mechanism Visualization: To further explain the source of Mul's advantage from a theoretical perspective, this section employs information plane analysis to quantify the dynamics of information flow during training. Experiments were conducted on the entire IP test set (10 089 samples). By extracting the output feature T_{attn} from the last CP-DMA module and the globally pooled feature T_{avg}, we observe their evolutionary trajectories in the tradeoff between "redundancy compression $I(X;T)$" and "discriminative information preservation $I(Y;T)$."

As shown in Figs. 9(b) and (c). Overall, the information evolution of all strategies at both feature layers adheres to the classic IB principle of "fitting followed by compression." In particular, through longitudinal cross-layer comparison, the Mul strategy exhibits a unique information filtering logic. At the T_{attn} layer, both $I(X;T_{\text{attn}}) \approx 0.17$ and $I(Y;T_{\text{attn}}) \approx 0.23$ are lower than those of the linear fusion strategies, indicating that it imposes stricter representational constraints at this stage. However, at the T_{avg} layer, the Mul strategy achieves the lowest redundancy ($I(X;T_{\text{avg}}) \approx 0.10$) and the highest discriminative information preservation ($I(Y;T_{\text{avg}}) \approx 0.81$). This demonstrates that Mul does not discard information randomly in the early stage; instead, it selectively filters out spatial background noise irrelevant to the target task through its second-order nonlinear gating mechanism. This early "purification" process provides a higher-purity input for the subsequent precise integration of global features, enabling the final representation to more closely approximate the "minimal sufficient statistic" in information theory. In contrast, due to the lack of effective early-stage modulation, the Add and Concat strategies allow more redundant information to propagate to subsequent layers, making it difficult for them to achieve an optimal information tradeoff at the global representation level. This fundamentally explains why they are inferior to multiplicative fusion in achieving deep feature interaction.

F. Comparison of Attention Mechanisms

1) Comparison With Standard Self-Attention: In this study, we construct four comparative configurations by introducing either the standard self-attention (S) [64] or our proposed central-pixel attention (C) in the spatial (Spa) and spectral (Spe) dimensions, respectively: SpaS–SpeS, SpaC–SpeC, SpaC–SpeS, and SpaS–SpeC. As shown in Fig. 10(a), all

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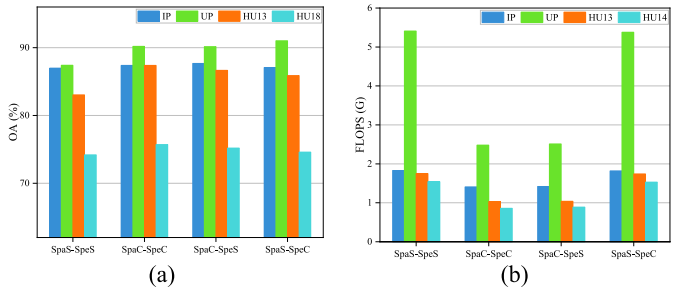


Fig. 10. Comparison between central-pixel attention and standard self-attention. (a) OA. (b) FLOPs.

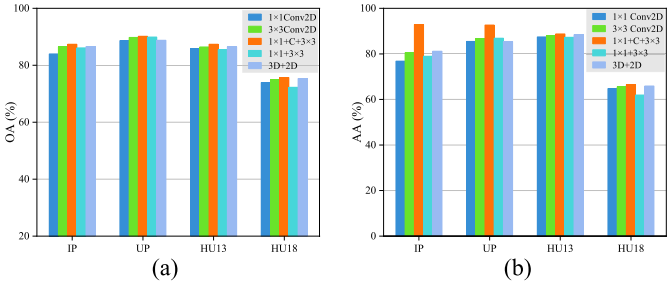


Fig. 11. Ablation study on the SSFE. (a) OA. (b) AA.

TABLE VII

COMPARISON OF DIFFERENT ATTENTION MECHANISMS IN SPATIAL AND SPECTRAL DOMAINS ON THE HU18 DATASET

Dataset	Spatial Attention	Spectral Attention	OA	AA	κ
HU18	✓	✓	**75.72±2.10**	**66.45±2.49**	**69.84±2.35**
	MHSA [64]	✓	74.87±2.45	65.58±2.46	68.97±2.26
	CBAM-SAM [65]	✓	74.16±2.27	65.28±2.46	68.59±2.37
	CVSSN-SVSS [25]	✓	74.81±2.34	65.46±2.51	68.62±2.40
	DBSSAB-CGSA [56]	✓	75.09±2.36	65.77±2.41	68.44±2.45
	CP-Transformer-CFA [28]	✓	75.34±2.25	65.79±2.44	68.83±2.26
	✓	MHSA [64]	75.46±2.27	66.08±2.36	69.19±2.37
	✓	SE [66]	73.28±2.38	63.79±2.41	67.14±2.53
	✓	CBAM-CAM [65]	74.87±2.45	65.58±2.46	68.97±2.26
	✓	SSSAN-SEA [67]	74.93±2.34	65.76±2.20	69.02±2.40
	✓	CP-Transformer-SSFA [28]	75.22±2.21	66.12±2.27	69.27±2.39

configurations incorporating central-pixel attention outperform the baseline (SpaS–SpeS), which exclusively uses standard self-attention. In particular, SpaS–SpeC achieves the highest OA on the UP dataset, while SpaC–SpeS delivers the best performance on the IP dataset. Our complete configuration, SpaC–SpeC, demonstrates significant superiority on the HU13 and HU18 datasets, with its performance on IP and UP being only marginally lower than their respective optimal configurations. More importantly, the FLOPs analysis in Fig. 10(b) reveals that a substantial reduction in computational cost is achieved as long as central-pixel attention is adopted in the spatial branch (i.e., SpaC). This is attributed to the elimination of redundant connections by the "one-to-all" attention mechanism. In summary, SpaC–SpeC achieves an optimal tradeoff between accuracy and complexity by maintaining high accuracy while significantly enhancing efficiency.

2) Comparison With Mainstream Attention Modules: We further replace the spatial and spectral attention modules of CP-DMA with current mainstream attention mechanisms and evaluate their classification performance on the HU18 dataset. In particular, the spatial branch is replaced with: the adaptive weight addition based spectral vector self-similarity (AWA-SVSS) module from CVSSN [25], the cosine and Gaussian–Euclidean similarity self-attention (CGSA) module from DBSSAB [56], the central-pixel focus attention (CFA) module from CP-Transformer [28], the standard MHA based on global modeling, and the CBAM [65] SA module (SAM). For the spectral branch, we employ the spectral self-attention (SSA) module from SSSAN [67], the spectral saliency focus attention (SSFA) module from CP-Transformer, MHA, the Squeeze-and-Excitation (SE) [66] attention module, and the CBAM channel attention module (CAM). As presented in Table VII, the model achieves the optimal OA of 75.72% when both our proposed spatial and spectral attention modules are used simultaneously. Replacing either branch with existing methods leads to a performance drop, which validates the effectiveness and superiority of our proposed mechanisms. It is noteworthy that modules also based on central pixels, such as CFA and SSFA from CP-Transformer and CGSA from DBSSAB, have achieved competitive results. By explicitly modeling the correlation between the central pixel and its neighborhood, these methods significantly outperform global attention mechanisms that lack the central-pixel prior. This further corroborates the necessity and effectiveness of incorporating the structural prior of the central pixel in patch-based HSIC tasks.

G. Ablation Studies

1) Ablation on Embedding Modules: To evaluate the effectiveness of the proposed SSFE module, this study compares five distinct embedding configurations: Configuration 1 (a single 1×1 2-D convolution), Configuration 2 (a single 3×3 2-D convolution), Configuration 3 (the three-stage scheme proposed in this work), Configuration 4 (the $1 \times 1 + 3 \times 3$ 2-D convolution without concatenation), and Configuration 5 (the cascaded scheme of $3 \times 3 \times 3$ 3-D convolution and 3×3 2-D convolution proposed in [17], [45], [48]). Experimental results presented in Fig. 11 clearly demonstrate that the proposed three-stage embedding scheme (Configuration 3) achieves superior performance across all tested datasets. A comparison with other configurations reveals that Configuration 1 yields the poorest performance, whereas the integration of 3×3 2-D/3-D convolutions (Configurations 4/5) or the standalone adoption of 3×3 convolution leads to a notable performance improvement. This finding indicates that capturing local spatial contextual information is crucial for enhancing the classification accuracy of HSIs. Notably, a comparative analysis between Configurations 3 and 4 (with the concatenation operation removed) shows that the latter exhibits a significant performance degradation across all datasets. A plausible explanation is that the concatenation operation facilitates the preservation and enhancement of multiscale information inherent in the original spatial–spectral features, thereby boosting classification accuracy.

2) Module Ablation: This section conducts ablation studies to analyze the role of each component, including the SSFE, the spatial MHA [incorporating SA and Gaussian positional encoding (G)], and the spectral MHA [incorporating depth-wise convolution (DWC), channel attention (CA), and

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TABLE VIII
IMPACT OF EACH MODULE ON CLASSIFICATION PERFORMANCE

Dataset	SSFE	Spatial		Spectral			Metrics		
		SA	G	DWC	CA	S	OA	AA	$\mathcal{K}$
IP	✓	✓	✓	✓	✓	✓	**87.38±2.11**	**92.85±0.84**	**85.73±2.35**
	Conv2D	✓	✓	✓	✓	✓	86.66±2.44	92.52±1.02	84.92±2.71
	✓	✗	✗	✓	✓	✓	86.91±2.29	92.41±0.98	85.12±2.53
	✓	✓	✓	✗	✗	✗	86.72±2.06	92.33±1.02	84.99±2.28
	✓	✓	APE	✓	✓	✓	85.26±2.67	90.67±1.40	83.72±2.93
	✓	✓	✓	✗	✓	✓	86.95±2.05	92.25±1.00	85.11±2.31
	✓	✓	✗	✓	✓	✗	87.03±2.27	92.61±0.95	85.31±2.40
UP	✓	✓	✓	✓	✓	✓	**90.20±3.55**	**92.64±1.91**	**87.30±4.14**
	Conv2D	✓	✓	✓	✓	✓	88.42±2.98	91.12±1.68	85.01±3.72
	✓	✗	✗	✓	✓	✓	87.79±2.84	90.82±1.26	84.21±3.49
	✓	✓	✓	✗	✗	✗	87.49±3.52	90.51±1.64	83.90±4.28
	✓	✓	APE	✓	✓	✓	86.52±3.79	89.17±2.06	82.61±4.37
	✓	✓	✓	✗	✓	✓	88.95±3.20	91.07±1.50	85.25±3.85
	✓	✓	✗	✓	✓	✗	89.27±3.16	91.84±1.45	86.80±3.77
HU13	✓	✓	✓	✓	✓	✓	**87.38±1.16**	**88.76±0.65**	**86.37±1.25**
	Conv2D	✓	✓	✓	✓	✓	86.58±2.15	88.15±1.70	85.52±2.32
	✓	✗	✗	✓	✓	✓	85.50±1.59	87.18±1.48	84.33±1.71
	✓	✓	✓	✗	✗	✗	85.67±1.51	87.38±1.06	84.52±1.63
	✓	✓	APE	✓	✓	✓	83.29±2.25	85.71±1.83	82.69±2.14
	✓	✓	✓	✗	✓	✓	86.34±1.45	87.45±1.07	85.60±1.55
	✓	✓	✗	✓	✓	✗	86.92±1.50	88.24±1.05	85.75±1.60
HU18	✓	✓	✓	✓	✓	✓	**75.72±2.10**	**66.45±2.49**	**69.84±2.35**
	Conv2D	✓	✓	✓	✓	✓	73.51±2.34	64.46±2.20	67.62±2.40
	✓	✗	✗	✓	✓	✓	74.36±2.27	65.28±2.36	68.59±2.37
	✓	✓	✓	✗	✗	✗	74.09±2.38	64.77±2.41	68.14±2.53
	✓	✓	APE	✓	✓	✓	73.06±2.51	63.82±2.64	67.25±2.49
	✓	✓	✓	✗	✓	✓	74.87±2.45	65.58±2.46	68.97±2.26
	✓	✓	✗	✓	✓	✗	75.22±2.21	65.92±2.27	69.23±2.39

spectral positional prior (S)], in influencing the classification performance.

The experimental results are shown in Table VIII. Removing any of the core components leads to a performance drop on all four datasets, which preliminarily confirms the necessity of each module design. In particular, when the SSFE module is replaced with a standard 3×3 2-D convolution, the OA on the UP dataset shows the most significant decline (dropping by 1.78%). This indicates that SSFE preserves more discriminative features through its "convolution–concatenation" structure during the embedding stage, providing a high-quality feature foundation for subsequent modules. In the spatial path, replacing G with absolute positional encoding (APE) results in a substantial decrease in OA across the four datasets by 2.12%, 3.68%, 4.09%, and 2.66%, respectively. This is because APE only provides absolute positional information and lacks association with the semantic features of the central pixel. In contrast, the "center-distance decay" physical prior introduced by G complements the SA that takes the central pixel as the query, enabling the model to focus on the core neighborhood and effectively filter out background noise from the edges. In the spectral path, removing the entire spectral module causes a 2.71% OA drop on the UP dataset, demonstrating the importance of spectral dimension modeling. Further removal of DWC weakens the model's ability to capture local spatial dependencies, leading to a 1.25% decrease in the UP dataset. Notably, performance also declines to varying degrees when both G and S are removed, indicating that G and S play key guiding roles in the spatial and spectral dimensions, respectively, helping the model efficiently extract discriminative information from complex features.

V. CONCLUSION

Inspired by the prior knowledge that "the central pixel dominates class semantics," this article proposes a CPFormer for HSIC, with its core innovation residing in the designed CP-DMA. A key distinction from existing studies lies in the fact that both the spatial and spectral attention mechanisms of the CP-DMA are directly guided by the central pixel, enabling deep interaction of orthogonal dimensional information through nonlinear element-wise multiplication. Combined with the proposed three-stage SSFE, the model's feature representation capability is significantly enhanced. Extensive experiments conducted on four public hyperspectral datasets fully validate the effectiveness of the proposed method.

However, CPFormer still has room for improvement: the central-pixel guided spatial–spectral attention mechanism is prone to modeling biases when handling class boundaries or slender linear ground objects (e.g., roads and rivers), stemming from the fact that the critical contextual information of such objects often exceeds the local neighborhood range. Future research will focus on optimizing the contextual modeling capability of spatial–spectral attention and exploring more efficient feature fusion strategies to further improve the model's classification performance in complex scenarios.

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Haobo Jiang received the B.Eng. degree from Nanchang University, Nanchang, Jiangxi, China, in 2022. He is currently pursuing the M.S. degree in computer science and technology with South–Central Minzu University, Wuhan, China.

His research interests include hyperspectral image processing, machine learning, and computer vision.



Qiankun Gao received the B.Eng. degree from Anhui Sanlian University, Hefei, Anhui, China, in 2023. He is currently pursuing the M.S. degree in computer technology with South–Central Minzu University, Wuhan, China.

His research interests include hyperspectral image processing, deep learning, and computer vision.



Haobing Chen received the B.Eng. degree from Zhengzhou University, Zhengzhou, Henan, China, in 2020, and the M.S. degree in computer science and technology from South–Central Minzu University, Wuhan, China, in 2024.

His research interests include hyperspectral image classification, deep learning, and computer vision.



Bin Zhou received the B.S. and M.S. degrees from the Department of Computer Science, National University of Defense Technology, Changsha, Hunan, China, in 1994 and 2002, respectively, and the Ph.D. degree from the School of Computer Science, Huazhong University of Science and Technology, Wuhan, China, in 2013.

He is currently a Full Professor with the School of Computer Science, South–Central Minzu University, Wuhan. His research interests include big data processing, computer vision, and intelligent data mining.

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